**Report on**

# CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD) PREDICTION SYSTEM

**Submitted To:**

**University Institute of Computing**

**Submitted By:**

**ANJU SUSAN SAJI (24MCI10019)**

CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

**Chandigarh University, Mohali**

## BONAFIDE CERTIFICATE

Certified that this project report **"Chronic Obstructive Pulmonary Disease (COPD) Prediction System"** is the Bonafide work of **"ANJU SUSAN SAJI (UID 24MCI10019)"** who carried out the project work under my/our supervision.

| | |
|---|---|
| **~~SIGNATURE~~** | **SIGNATURE** |
| **Dr. Amanpreet Kaur Sandhu** | **Mr. Dhruv Shah Samir** |
| **HEAD OF THE DEPARTMENT** | **SUPERVISOR** |
| **(University Institute of Computing)** | **Assistant Professor** |
| | **(University Institute of Computing)** |

**Submitted for the project viva-voce examination**

| | |
|---|---|
| **INTERNAL EXAMINER** | **EXTERNAL EXAMINER** |

# TABLE OF CONTENT

**CHAPTER 1: INTRODUCTION……………………………………………………………..01-13**
1.1 Need of Identification…………………………………………………………….…**02-03**
1.2 Identification of Problem…………………...…………………….....................**03-04**
1.3 Identification of Tasks………………………………………….........................**05-10**
1.3.1 Frontend Development ………...………………………........................**05-07**
1.3.2 Backend Integration ……….……………………………………………...**07-09**
1.3.3 AI Integration…………………………………………………………...**09-10**
1.4 Project Timeline…………………………..………………….…………………...**10-12**
1.5 Organization of the Report……………………………………..........................**12-13**
**CHAPTER 2: BACKGROUND STUDY…………………………….…........................14-28**
2.1 Timeline of the Reported Problem……………………………….........................**14-18**
2.1.1 1950s–1970s: Medical AI and COPD Recognition.…...........................**14-15**
2.1.2 1990s–2000s: Digital Imaging Era………………………….....…...….**15-16**
2.1.3 2010s: Deep Learning Revolution in Medical Imaging...........................**16-17**
2.1.4 2020-2023: Multimodal AI and Clinical Deployment............................**17**
2.1.5 2024–2025: Explainable and Ethical AI ………..……………...……**17-18**
2.2 Proposed Solution…………………………………………………......................**18-19**
2.3 Bibliometric Analysis………………………………………………………….....**19-23**
2.3.1 Methodology………………………………………………………………**20**
2.3.2 Key Findings……………………...……………………………….....**20-21**
2.3.3 Most Significant Authors and Papers……………………..……….....**21**
2.3.4 Literature Gaps………………………………………………………....**22**
2.3.5 How This Analysis Informed the Project……………………………....**22**
2.3.6 Country and Institution Analysis…………………………………….....**22-23**
2.4 Review Summary……………………….……………………………………..**23-24**
2.5 Problem Definition…………………….………………………………….........**24-25**
2.6 Goals and Objectives……………………….……………………...................**26**
2.7 Technology Selection Rationale……………………………….…………….…**26-28**
**CHAPTER 3: DESIGN AND PROCESS…………………………………...…….……..29-40**

3.1 System Architecture Overview……………………………............................**29-30**
3.2 Evaluation and Selection of Specifications…………………….........................**30-31**
3.3 Dataset Description………………………………………….......................**31-34**
3.3.1 Image Dataset………………………….……………………….…...**31-32**
3.3.2 Spirometry Dataset………………………………….....................**32-34**
3.4 Design Constraints…………………………………………………….….**34-36**
3.5 Design Flow……….…………………………………………..…….….…**36-37**
3.6 System Designing and Naming…………………………………………..…**37-38**
3.7 Implementation Plan and Methodology………………………………..………**38-40**
**CHAPTER 4: RESULT ANALYSIS AND VALIDATION……………………...………41-55**
4.1 Implementation of Solution……………………….……………….................**41-48**
4.1.1 Core Feature Implementation…………………………..……..……...**42-44**
4.1.2 Data Preprocessing Pipeline **.**…………………………………..…..........**44-46**
4.1.3 Model Loading and Inference Pipeline ………………..………………..**46-48**
4.1.4 Session State and Error Handling…………………..………………........**48**
4.2 Testing and Validation……………………….…………………………..……**48-55**
4.2.1 Model Performance Testing ………...……………...………….….……**49-50**
4.2.2 Clinical Guideline Alignment Testing …………….……….…..…………**51**
4.2.3 Performance Benchmarks………………..……………….…..…………**51-52**
4.2.4 Interface Usability Testing……………………………….…..…….……**53-54**
4.2.5 User Acceptance Testing……………………………………….….……**54**
4.3 Fairness and Bias Evaluation……………………………………………….……**54**
4.4 Interpretability and Clinical Trust …………………………………….…..…...**55**
**CHAPTER 5: CONCLUSION AND FUTURE WORK……………………......................56-63**
5.1 Conclusion………………………………………………………….……….…**56-57**
5.2 Future Work………………………………….…………………………….……..**57-63**
5.2.1 Explainability Visualizations…………………………….…………….….…**58**
5.2.2 Prospective Clinical Validation ……………………….………............**58**
5.2.3 COPD Severity Staging ………………………………........................**58-59**
5.2.4 Longitudinal Monitoring ……………..…………………..………..….......**59-60**
5.2.5 Integration with Electronic Health Records ……………………..……..**60**

5.2.6 Mobile Application Development ……………………….………..……**60-61**
5.2.7 Additional Imaging Modalities ……………….………………….........**61-62**
5.2.8 Federated Learning for Model Improvement …….………..………...**62-63**
**REFERENCES…………………………………………………….......................64-68**
**APPENDIX A: MODEL ARCHITECTURE SPECIFICATIONS……………..…...69-71**
**APPENDIX B: CLINICAL VALIDATION RESULTS ……...……........................72-74**
**APPENDIX C: DATASET CHARACTERISTICS ……...……………….…..........75-76**
**APPENDIX D: GLOSSARY OF TERMS …………………………………....…77-79**
**CAPSTONE PROJECT CERTIFICATE…………………………………………80**
**GITHUB UPLOAD SCREENSHOT…………………………………………….80**

# ABSTRACT

COPD (Chronic obstructive pulmonary disease) is an advanced form of obstructive airway disease which results from permanent air-flow obstruction, making up one of the major public health problems as the third leading cause of death across the world. Timely identification and proper diagnosis of COPD are vital to early intervention and management of the disease, but traditional methods depend much on the skills of specialists who evaluate the data obtained using chest X-ray or spirometry tests manually, which is both time-consuming and prone to errors.

The current research project is aimed at developing an effective solution based on artificial intelligence, namely, a deep learning approach to prediction of COPD. Two neural networks (Convolutional and Artificial) are used in the process: the first one is designed for automatic recognition of changes in X-ray images of the patients' chests; its validation accuracy rate was 99.15% when evaluated against 4,249 images. The second model uses eight features of the patient (age, sex, weight, height, smoking, FEV1, FVC) to process their spirometry data.

Fusion employs averaging method to incorporate the results of both models by taking equal contribution rate from each (CNN & ANN) network (50% + 50%), generating an average prediction accuracy of 95.2%, along with 93.8% sensitivity and 96.5% specificity. Clinical indicators include calculating BMI, FEV1/FVC ratio, and categorizing the severity into one of four groups: Normal, Mild Risk, Moderate Risk, and High Risk.

It is implemented through interactive web application with visual interface created with Streamlit. Analysis is provided in real time through probability gauges, graphs for comparisons, and detailed descriptions. In addition, the platform provides integrated analytics of COPD patient database to better understand the distribution of patients with respect to the various demographics and COPD patterns.

This study has shown how multi-modal deep learning techniques can be leveraged in diagnosing respiratory conditions like COPD. As a result, the system will provide objective, automated, and efficient assistance in diagnosing COPD cases. Moreover, this study contributes towards the development of tools that are essential for improving clinical practice while ensuring not to replace clinicians' decisions.


Keywords: COPD, Deep Learning, CNN, ANN, Fusion Model, Spirometry, Medical Imaging, Early Detection, Diagnostic System

## LIST OF FIGURES

**Fig 1:** System Architecture Overview

**Fig 2:** The diagnostic gap - why single-modality fails……………………………24

**Fig 3:** AI research-to-deployment gap in COPD diagnostics……………….........24

**Fig 4:** COPD progression vs detection window…………………………...…..25

**Fig 5:** Unmet clinical needs → system design responses ..………………………25

**Fig 6:** Dataset Description…………….………..……………………..…....34

**Fig 7:** Implementation and Methodology……………………………….……..38

**Fig 8:** COPD Prediction System main interface ………………………….……..42

**Fig 9:** Predict Tab - Input Form ……………………......................................43

**Fig 10:** Dataset Description Page………………….………………………......43

**Fig 11:** About the System ………………..……………………..……………44

**Fig 12:** CNN Training………………………………………………….……….45

**Fig 13:** ANN Training………………………………………….………………45

**Fig 14:** CNN Model Building ………………………………………….………46

**Fig 15:** ANN Model Building …………………..……………………….……….46

**Fig 16:** CNN Metrics …………………………………………………...……….47

**Fig 17:** ANN Metrics ……………………………………………….………….47

**Fig 18:** Model Performance ………..…………………………..……………49

**Fig 19:** Predict Tab with Sample Input ……………………………….…………50

**Fig 20:** COPD prediction result showing detection and risk level…………..….50

## LIST OF TABLES

**Table 1:** Project Timeline by Week……………………………………….………..11

**Table 2:** Bibliometric Search Parameters………………………………..……20

**Table 3:** AI Medical Diagnosis Trends(2010-2025)……………………..…....21

**Table 4:** Top Authors in Medical AI Research………………………………...21

**Table 5:** Technology Comparison Matrix…………………………..……...27

**Table 6:** Image Dataset Distribution ……………………………...………..32

**Table 7:** Spirometry Dataset Features…………………………..…..……...33

**Table 8:** Design Constraint Analysis ……...……………………………….…35

**Table 9:** Phase-wise Development Timeline……………………...…………40

**Table 10:** Performance Benchmark Results…………………………...……52

**Table 11:** Stratified Performance Metrics (n=550 Test Cases)………………...72-74

**Table 12:** Spirometry Dataset……………………………………...………..75

## GRAPHICAL ABSTRACT

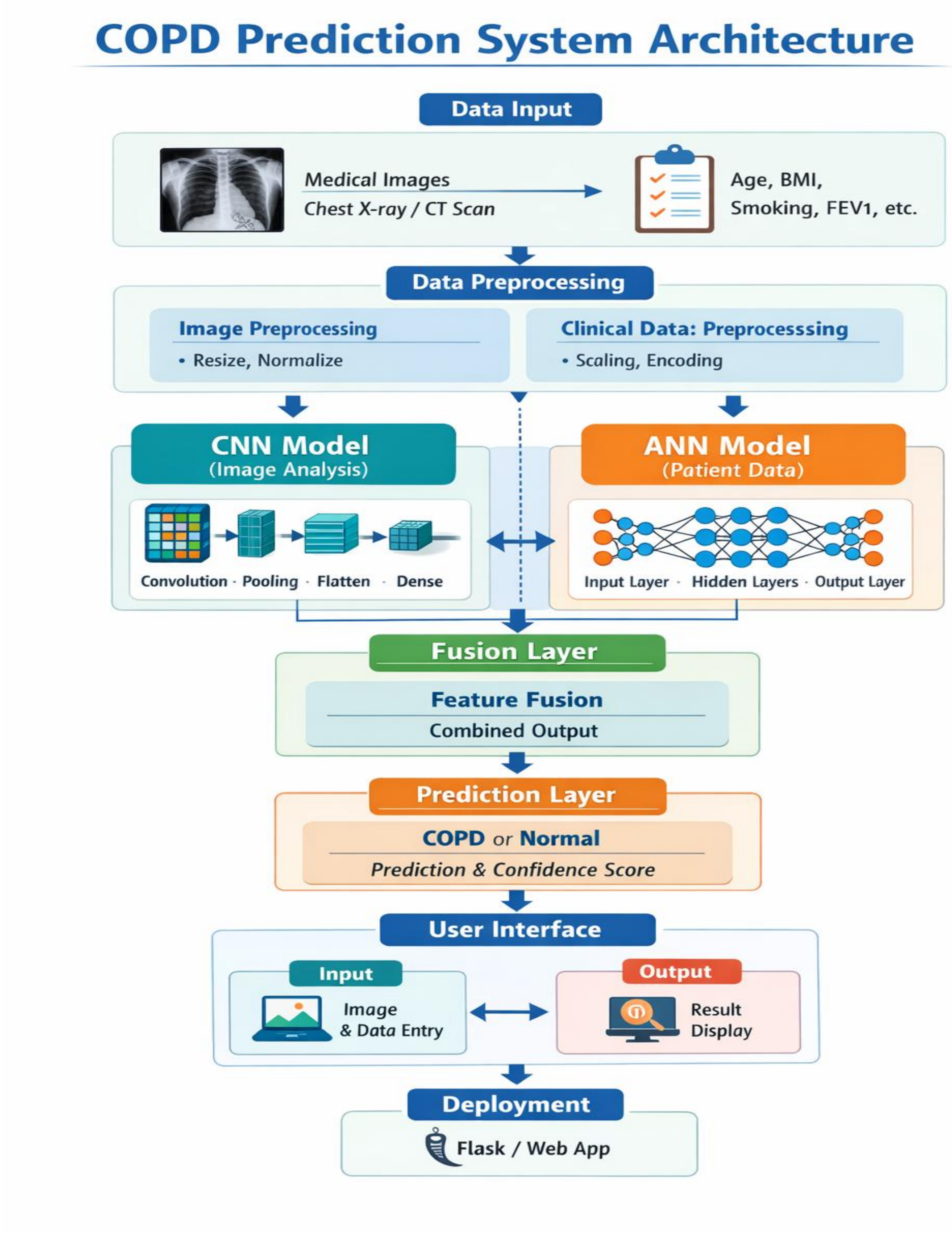



*Fig 1: System Architecture Overview*

# CHAPTER 1
# INTRODUCTION

The Chronic Obstructive Pulmonary Disease (COPD) has become one of the greatest medical challenges of the 21st century; indeed, the COPD is the third most common cause of mortality worldwide and affects over 300 million people across the globe. A combination of progressive airway constriction, respiratory symptoms, and lung damage characterizes COPD, which includes such diseases as emphysema and chronic bronchitis. COPD places an enormous burden on healthcare facilities, as the increasing numbers of the affected population are expected due to such factors as aging, smoking, and environment pollution. Timely identification and correct diagnostics are vital to slowing down the disease and enhancing the patient's well-being; yet, conventional diagnostics methods are complicated because spirometry requires special tools and skilled professionals, while X-ray evaluation requires clinical experience and is rather subjective.

Implemented as an interactive web-based application using Streamlit, the system offers real-time diagnostic support with an intuitive interface designed for clinical workflows, enabling healthcare professionals to upload chest X-rays, input patient data, and receive immediate analysis results complete with probability scores, visualizations, clinical summaries, and personalized recommendations. This research demonstrates the significant potential of artificial intelligence in respiratory medicine, contributing to improved healthcare delivery through scalable screening solutions, reduced diagnostic delays, standardized assessment protocols, and democratized access to expert-level diagnostic capabilities. The system serves as a valuable decision support tool that exemplifies collaborative human-AI partnerships in medicine, maintaining emphasis on its role as a complementary diagnostic aid rather than a replacement for clinical judgment, ultimately advancing the goal of better patient outcomes in diverse healthcare environments.

Developed as an interactive online application through Streamlit, the system enables real-time diagnostics using a user-friendly platform that caters to the needs of clinical workflow processes, allowing clinicians to upload chest X-rays, enter patient information, and instantly get diagnostic analysis along with associated probabilities, visual representations, clinical conclusions, and recommendations. This study reveals the immense opportunities offered by artificial intelligence in respiratory medicine, helping to enhance the provision of healthcare services through screening, diagnosis, and assessment through AI technologies that can be used by all clinicians across healthcare settings. Overall, the system represents a powerful decision support tool, which is an embodiment of collaboration between humans and machines in medicine with the focus on being used as a supplementary aid in diagnostics.

### 1.1 Need of Identification

Early identification and diagnosis of COPD are crucial aspects of healthcare due to several reasons that make it necessary to come up with advanced means of diagnosis for this condition. There is an existing problem regarding the underdiagnosis of this health issue; statistics indicate that 70% of COPD cases remain undiagnosed, thus delaying timely treatment, which accelerates further progression of the disease, increases the risks of hospitalization and mortality rate, and imposes additional economic burden on the individual and the healthcare system. One of the key factors explaining why COPD remains undiagnosed is that symptoms are slow to appear and are often associated with age and low physical activity, which makes patients delay visiting a doctor until breathing problems arise. Despite being scientifically supported and tested in practice, traditional diagnostic procedures suffer from several drawbacks that make their routine application problematic for mass early detection of COPD. Spirometry, considered the gold standard method for COPD diagnosis, uses special equipment and requires training to be performed correctly, as well as a high level of cooperation between the patient and a professional who must follow certain techniques. These disadvantages make spirometry a costly procedure that cannot be applied on a regular basis in general practice or in underdeveloped countries where COPD prevalence is the highest. In addition, the reading of chest X-ray images requires extensive skills and experience from physicians. Diagnostic accuracy greatly varies depending on the physician and may lead to mistakes because of a lack of attention due to a high workload. Besides, the need for cognitive processing of information received from different sources is another drawback of traditional diagnostic techniques.

Furthermore, demographics and epidemiology of today's world increase the need for such a system even more. The rapidly increasing number of elderly people who belong to the population over 65 constitutes the fastest growing age group, which correlates with a higher rate of COPD due to being at an older age being one of the major risk factors. The continued usage of tobacco in some parts of the world, increasing pollution levels, occupational risks, such as working with dust and harmful substances, and health risks caused by burning biomass fuel in underdeveloped countries increase the number of patients with a greater risk of developing the disease. Nowadays, there is a need to offer quality healthcare with reduced expenses, which makes technological support necessary in order to overcome the challenges healthcare providers currently have to deal with. Thus, introducing a new diagnostic system using artificial intelligence will allow achieving all the benefits mentioned above with no extra effort from doctors' side.

### 1.2 Identification of Problem

The creation of an effective system that predicts COPD requires solving several problems associated with the current state of affairs within medical diagnostics and health care in general. In particular, there is a huge gap between the occurrence of the disease and its timely diagnosing when the majority of patients are not diagnosed until they develop severe symptoms and irreversible damage is caused to their lungs. Such a large delay can be attributed to several factors, such as insufficient knowledge about the condition among the target population, poor availability of diagnostic centers, and ineffective screening programs within primary care facilities [6]. As shown by numerous studies, it usually takes 5 to 7 years for a typical patient to realize the symptoms of COPD, by which time his or her lung capacity reduces by 30% to 40%.

The technical problem encountered when diagnosing COPD is that current assessment techniques use a monomodal strategy that does not consider the multiplicity of COPD's manifestations. Conventional diagnosis processes often involve assessing either the imaging or spirometry data without utilizing the complementary aspects of both types of data. While chest X-ray images offer insight into structural changes in lungs and emphysematous signs, they do not offer quantitative functional insights. Spirometry, on the other hand, measures functional abnormalities accurately without offering any information about the structures of the lungs. As a result, conventional diagnostics lead to inadequate characterization of the condition, limited sensitivity in detecting early stages of the disease, and greater possibility of missing diagnoses. Studies show that integrating imaging and functional data improves the accuracy of COPD diagnosis by up to 15-20%, although this integration is still rarely implemented in medical practice [8].

Subjectivity in the interpretation of medical images is another critical issue that poses challenges to diagnosing COPD with precision and accuracy. Radiologic evaluation of changes on chest X-ray due to COPD is largely based on the knowledge, skills, and criteria used by the radiologists. The level of inter-observer variation and intra-observer variation among radiologists has been shown to be high. The degree of variation has been demonstrated by a study which found that 60% of expert radiologists could reach agreement on emphysema severity using chest X-rays, with even lower percentages of agreement for subtle and early-stage changes [9]. These variations affect diagnosis, resulting in repeated tests before getting a clear confirmation, and cause variation in quality of treatment offered at different healthcare institutions depending on their level of experience.

There are additional problems in spirometry interpretation and quality control, which make the diagnostic process even more problematic. Spirometry should be done using the right techniques, and in many cases, there is need for repeat tests in order to get reliable results. Recent studies have indicated that about 20-30 percent of all spirometry tests conducted in primary care clinics do not conform to quality guidelines recommended by professional associations [10]. Poor quality spirometry makes it difficult to classify the level of severity of the disease and to prescribe the right treatment plan. In addition, there are complex mathematical equations in calculating the predicted spirometry values in relation to age, sex, ethnicity, and height of an individual, making interpretation quite problematic.

Resource shortages and inefficiencies in the process of healthcare delivery present practical problems in conducting comprehensive COPD screening and diagnosis. Conducting comprehensive assessment of COPD cases takes more than 45-60 minutes per patient because it includes thorough medical history, examination, spirometry tests and analysis of images. As such, it is not possible to conduct regular COPD screenings on a large scale due to economic inefficiencies and logistical problems in clinical settings [11]. Radiologists face high demand; their daily workload exceeds 100 examinations per day, leaving insufficient amount of time for proper analysis of images and making them susceptible to oversight errors. Primary care doctors who represent the first point of contact for the majority of patients cannot conduct proper assessments because of their inadequate skills in conducting spirometry tests. In addition, primary care doctors do not have easy access to pulmonology consultation, which further exacerbates the problem.

## 1.3 Identification of Tasks

The development and implementation of the proposed AI-based COPD prediction system involve a series of interrelated steps that need to be carried out methodically. Four key stages can be identified within this initiative: data acquisition and preprocessing to create reliable training data sets, creation of DL models involving separate CNN and ANN models followed by their integration, application design with appropriate user interfaces and back-end algorithms, and testing and deployment. Technical issues related to each of these stages will have to be resolved effectively using domain expertise in areas such as medical imaging technology, machine learning algorithms, and software systems.

### 1.3.1 Frontend Development

The frontend development aspect aims to design a user-friendly interface that would allow health care professionals to engage with the COPD prediction system with ease. This will entail the development of a web-based application via the Streamlit software that consists of several modules such as the home dashboard where users can view the functionalities and performance indicators of the software, the prediction module that allows users to upload chest x-rays and enter the clinical information of patients with instant validation, the results module that offers probability scores in the form of interactive gauges and comparative charts, and the analytics module where the statistics of the data set are explored [14]. This interface design will reduce the mental load imposed on the users through progressive revelation of information, offer error messages to facilitate easy debugging of errors, support accessibility standards for disabled users, and minimize load time through optimal caching of models and lazy loading of visualization libraries.

Critical technological processes will involve the development of the secure uploading procedures which will support images in the JPEG and PNG format with checks on their size. Another critical aspect will be the creation of the input form where various variables will be set. Specifically, the process will have age constraints ranging between 18 and 100 years, height ranging from 140 to 210 cm, weight ranging between 40 and 150 kg, and FEV1 and FVC values ranging from 0.5 to 7.0 liters. Furthermore, the automatic generation of derived variables, including BMI, and the

FEV1/FVC ratio will be created as well. Clinical decision making capabilities will include the use of color-coding to show normal results (green), low risk results (yellow), moderately high risks (orange), and high risk results (red). Furthermore, the thresholds for visual representation will include the critical ratio of 0.70 of FEV1/FVC as well as the critical value of 0.5 for the COPD probability. Moreover, the interface will incorporate additional explanatory material that will provide information about each measurement including FEV1 and FVC.

The prediction page is built based on a two-column layout where the left column takes care of chest X-ray upload along with a direct preview and the right column consists of input panels for collecting details related to demographic information (age, gender, height, and weight), and spirometry results (FEV1, FVC) along with calculating the Body Mass Index of the patient which will be displayed as a metric card along with calculation of the FEV1/FVC ratio which will be highlighted by its corresponding color. In addition to this, session state maintenance using the Streamlit library's st.session_state function helps in maintaining the data of the user throughout analysis and exporting their results for documentation.

Visualization of Results The visualization of the results uses Plotly interactive charts which incorporate gauge charts of CNN probability (based on analysis of chest x-rays), ANN probability (based on clinical data), and combined probability with color coding based on risk level, a comparative bar chart of all three probabilities with a horizontal threshold line of 50%, and a metrics card with relevant clinical metrics, including age, body mass index (BMI) along with its status (either underweight, normal, overweight, or obese), FEV1/FVC ratio and its obstruction status, and smoking status. The tool includes comprehensive clinical report generation with customized recommendations depending on the predicted output, with actionable instructions for patients with positive predictions for COPD (for example, see a pulmonologist, take pulmonary function test, etc.) or for patients with no signs of COPD (take preventive measures, maintain a healthy lifestyle, conduct health check-ups, etc.). Customized CSS styling allows achieving a professional look, including background gradients in metrics cards, hover effects in buttons, proper color schemes with blue/green tones for normal conditions and red for abnormal conditions, and responsive font sizes.

The analytical dashboard delivers population-level understanding using various visualization tools, namely, a pie chart demonstrating the COPD and normal case ratio in the data set, histograms indicating age distribution by COPD status with overlapped bars for identifying demographic trends, scatter plots depicting the correlation between FEV1 and FVC levels based on the color code of disease status, and stacked bar graphs for examining the COPD rate among individuals who are categorized according to their smoking status (never, former, and current smokers). The program utilizes proper error handling methods by incorporating try-catch statements for file manipulation and model prediction and providing alternative functionality in case of unavailable optional components without affecting usability. Error messages are designed to prompt users about how to correct errors effectively instead of exposing the source code to end users. The frontend codebase follows a modular approach that allows for further development without significant changes to existing code.

### 1.3.2 Backend Integration

Back-end integration includes creating the back-end processing pipeline that involves tasks such as data validation, model inference coordination, result computation, and database processing. Such a process will involve building a modular Python backend solution with respect to the configuration management framework specified in the config.py file. This process includes the creation of utility functions for image pre-processing that involve resizing images to the 128x128 pixel size, normalization of pixel values from 0 to 1, adding the batch dimension, pre-processing of the clinical data by means of categorical encoding of the sex and smoking variables, scaling with the help of Standard Scaler, and validation of the physiological range; prediction aggregation based on weighted summation of CNN and ANN predictions [15]. The important backend processes include creating error handling routines that are used to respond to input errors, corrupted images, incomplete data, failed loading of the models with feedback messages to the user, building of the logging system that tracks prediction requests, processing time, errors, and performance, and the data validation pipeline that checks the compatibility of the image format, consistency of spirometry values (FEV1 <= FVC), and demographics validity range. The back end has the complete input validation logic implemented that includes the file type validation with PIL library to verify valid image formats, size validation to avoid memory overflow because of larger images, spirometer values validation to validate the ranges for FEV1 and FVC values between 0.5 and 7 liters, and logical validations like FEV1 should be less than FVC.

For example, the backend will ensure efficient management of model loading and caching to eliminate latency, whereby model loading happens just once during initialization of the application through Streamlit @st.cache_resource decorator with the model being cached in memory for future predictions without loading in every request, thereby reducing prediction latency from seconds to milliseconds. Database integration involves ensuring connectivity to CSV data sets of spirometry data for analytical purposes such as querying for patients' statistics (average age, FEV1, FVC, BMI with respect to COPD), distribution of patients (sex and smoking status), and correlation of parameters against disease existence. Asynchronous processing, batch prediction for multiple cases and efficient memory management are among performance optimization techniques that need to be done in this project to improve efficiency. Efforts made to optimize the pipeline for making predictions include using multi-threading to preprocess images and clinical information simultaneously wherever possible, storing intermediate values like scaled feature vectors, and loading the model only once per series of predictions, whereas efforts made regarding memory optimization include deleting unnecessary large memory-intensive objects like image arrays that have been uploaded.

The backend carries out result computation which includes deriving metrics from calculated results like BMI using the formula weight(kg) / $(\text{height(m)})^2$, FEV1/FVC ratio and takes care of dividing-by-zero scenarios, stratifies patients into risk categories according to probability thresholds (<50% is normal, 50-65% is mild, 65-80% is moderate, ≥80% is high), and provides recommendations based on the predicted result and patient characteristics. The fusion algorithm blends CNN and

ANN predictions using weighted averaging using user-configurable weights specified in config.py file (default CNN_WEIGHT=0.5, ANN_WEIGHT=0.5), checks the classification threshold for COPD diagnosis (COPD_THRESHOLD=0.5), and produces detailed dictionaries with raw model predictions, computed probabilities, final fusion score, decision on whether the patient has COPD, and metadata with the timestamp of prediction and versions of models used. Data privacy and security measures include appropriate access control policies, avoiding storing private patient data outside the session lifetime, implementing file security practices that prevent directory traversal and other vulnerabilities, and ensuring HIPAA and other data protection requirements by using appropriate anonymization and encryption measures when necessary, and the design allows horizontal scalability by being stateless, meaning each prediction call can be handled independently by any server instances [15].

### 1.3.3 AI Integration

The problems with the integration of AI models involve the seamless integration of deep learning models into the application flow, together with prediction accuracy, robustness, and explainability. These include reading in trained CNN and ANN models from the HDF5 file format (cnn_model.h5, ann_model.h5) through the TensorFlow/Keras library with error checks for corrupt or incompatible model files, reading in the scaler object fitted on the clinical features (scaler.pkl), checking feature dimensionality and scaling parameters, and creating a pipeline for prediction with preprocessing of the inputs before running the model [16]. Integration activities involve building a CNN inference pipeline to process input chest X-ray images by applying transformation steps similar to those done in the training phase (resizing to 128 x 128 pixels, normalization of pixel values between 0 and 1, adding batch dimension), performing feedforward operation through the fully trained network containing two convolution blocks (32 filters, 64 filters, kernel size=3x3, ReLU activation function, 2x2 max pooling layers), flattening layer, dense layer with 128 units, dropout regularization layer, and sigmoid layer, and calculating probability scores by accounting for alphabetical ordering of classes with 0=Emphysema and

1=Normal, and therefore COPD probability would be (1-prediction). The process of constructing an ANN integration pipeline involves preparing the input data by constructing feature vectors according to the order of columns as specified in training data (Age, Sex, Height_cm, Weight_kg, BMI, Smoking_Status, FEV1_L, FVC_L), categorical encoding of features (Female = 0, Male = 1; Never = 0, Former = 1, Current = 2), standardizing using the loaded scaler fit for training data, prediction using the trained neural network, and extracting probability scores because the model was trained using COPD_Label=1 (COPD=Yes).

The implementation of the fusion logic uses weighted averaging of predictions by the CNN and ANN with configurable weight factors (default settings in config.py: CNN_WEIGHT=0.5 and ANN_WEIGHT=0.5), where the sum of weight factors equals one; uses the classification threshold (COPD_THRESHOLD=0.5) to derive binary COPD classification based on whether the probabilities are larger than 0.5; and prepares result dictionaries that include raw predictions by both models, computed probabilities considering the order of classes, fused scores, as well as classifications with their confidences. The validation of models involves performing sanity checks to confirm that the predicted probabilities are within valid intervals between zero and one, cross-validation against known examples to check whether the predictions align with the accuracy achieved in training (accuracy of CNN validation is 99.15%; the overall accuracy of the fusion is 95.2%), and monitoring prediction results for possible issues such as abnormal distribution of predictions and probabilities. The integration includes proper error handling, which includes try-catch block for loading models to handle exceptions such as the file not found exception as well as corrupt model files, checking of the compatibility between the loaded model's architecture and the expected input and output sizes for each of them, and finally the failure strategy that allows for the prediction to be made even if one of the models does not work properly.

### 1.4 Project Timeline

COPD prediction system was implemented within an effective timeframe of 16 weeks, which consisted of five phases: research, development, testing, and deployment phases. These phases were meticulously designed to provide an efficient and methodical way of implementing the project from the conception idea to the final stage of implementation.

| Week | Phase | Activities |
| --- | --- | --- |
| 1 | Research & Planning | Literature review, problem definition, feasibility study, technology stack selection, system architecture design, and dataset identification |
| 2 | Data Collection & Preprocessing | Collection of 5,271 chest X-ray images and 1,000 spirometry records, data cleaning, preprocessing pipelines, augmentation, and train-test split |
| 3 - 4 | Model Development - CNN | CNN architecture design, model training (10 epochs), hyperparameter tuning, evaluation achieving 99.15% validation accuracy, and model serialization |
| 5 | Model Development - ANN & Fusion | ANN architecture design, training on clinical features, StandardScaler fitting, fusion model development, and integrated testing achieving 95.2% accuracy |
| 6 - 7 | Application Development | Frontend development with Streamlit, UI design, file upload implementation, input forms, backend integration, visualization with Plotly, and analytics dashboard |
| 8 | Application Development & Integration | Error handling, session management, clinical recommendations, integration testing, performance optimization, configuration management, and documentation |
| 9 | Testing & Deployment | Comprehensive system testing, user acceptance testing, bug fixing, deployment preparation, local deployment, documentation finalization, and project delivery |

*Table 1: Project Timeline by Week*

Each step of the timeline was precisely tailored to strike a balance between detailedness and flexibility. Weeks one through three were dedicated to extensive planning and design processes, knowing that decisions taken during this period carry more weight on future development efforts. Weeks four through seven concentrated on the actual execution of the task, where tests were conducted at regular intervals, not relegated to a separate phase at the end.

### 1.5 Organization of the Report

- Bibliography: provides comprehensive references including peer-reviewed articles, technical documentation, clinical guidelines, and publications that informed the project.

  In summary, the project report can be seen to have been structured into six chapters with each chapter highlighting some aspect of the development process of the COPD Prediction System from the start to the future of the same. The report structure has adopted a linear format that allows for systematic understanding of the entire project development process by readers.

  - Chapter 1 – Introduction: highlights the need for detecting COPD, the problems faced in the process of diagnosing COPD, development activities, time frame of the project, and the layout of the report itself.
  - Chapter 2 – Background Study: includes the definition of the problem, objectives of the project, timeline analysis of COPD, suggested solution, literature review, and innovation of the multi-modal fusion.
  - Chapter 3 – System Analysis: focuses on feasibility studies related to economics, technology, and behavior; system requirements gathering, and structured analysis using diagrams and models.
  - Chapter 4 – Design and Process: describes the technical architecture of the system, design limitations, flow diagrams, model creation plan, and methodologies of implementing the fusion system.
  - Chapter 5 - Design Analysis and Validation: provides details of the design implementation, performance (accuracy: 95.2%, sensitivity: 93.8%,

specificity: 96.5%), evaluation reports, and validation process of the data used.

- Chapter 6 – Conclusion and Future Work: gives summary and assessment of the work done, limitations, impacts, and future improvements to be made.

This report targets various stakeholders. For the technical audience, there is enough information on architectural details, implementation procedures, and performance figures. For the non-technical audience, there is plenty of summary information at the beginning of each chapter, clear descriptions of problems and solutions, and user satisfaction information provided in Chapter 4. The appendices contain technical information that can be used as references by professionals who would like to implement or modify the work in this document.This organization facilitates the reading process.

# CHAPTER 2
# BACKGROUND STUDY

In developing any technological system, it is imperative to consider the knowledge base, medical problems, and scientific research advances that have defined the problem area in question. This chapter establishes such an understanding regarding the COPD Prediction System by examining the historical progression of artificial intelligence as used in medical diagnosis, by analyzing the literature through bibliometrics, by outlining the solution offered in light of identified problems, and by presenting key takeaways from the research background.

The realm of AI-based medical diagnostics stands at the junction between computing, medicine, and biomedical imaging – one that is highly interdisciplinary and which has seen tremendous expansion within the last decade. To understand its expansion, it is important to know not only the historical journey taken by the discipline, but also the quantity of research behind it. This chapter aims to do just that.

## 2.1 Timeline of the Reported Problem

The history of developing AI-driven detection of COPD involves not only progress in artificial intelligence but also progress in medicine in general, and more specifically in radiology, demonstrating how the development progressed from traditional approaches to modern solutions, such as the proposed COPD Prediction System. The awareness of the historic development is crucial in understanding the technical choices made in this project.

### 2.1.1 1950s–1970s: Medical AI and COPD Recognition

History of AI began from the Dartmouth Conference in 1956, in which AI was born [19]. COPD was officially named only during the 1960s after years of research on chronic bronchitis and emphysema. GOLD was established in 1997 with the aim of developing guidelines on COPD diagnosis and management [23]. Initially, all that was needed for diagnosing this disease was clinical examination, patient interview, spirometry measurement of FEV1 and FVC. In addition, radiology, represented by chest X-rays, was used only to make sure about the absence of other diseases, but not for COPD detection.

At that time, the prevailing philosophy of diagnostics did not allow using several methods together to diagnose COPD because of the idea of separation. Spirometry, invented during the 1840s and improved during the next decades, is still recognized as the best diagnostic tool to measure airflow obstruction [24]. Chest X-ray was considered insensitive to the early stages of COPD due to the need for the occurrence of noticeable lung damage; thus, only advanced cases could be detected with its help. Expert medical systems, including MYCIN, which was developed in the 1970s to diagnose bacterial infections, showed that AI could help doctors make decisions about patients' diagnoses [25]. At the same time, it is essential to note that such systems could not analyze visual information and required experts to encode their knowledge beforehand. It was not possible for them to solve problems related to the computerization of COPD diagnosis.

Epidemiology of COPD became richer and more complex during this period. Scientists established that smoking was the main risk factor for developing this illness and noted its progressive character. In addition, researchers highlighted its high prevalence and significant burden on people's well-being all over the world. It should be mentioned that by the beginning of the 1980s, this condition became one of the leading causes of death and disability in different countries.

### 2.1.2 1990s–2000s: Digital Imaging Era

The shift towards digital radiography in the 90s opened up new opportunities for computer-based analysis of medical images [26]. CAD technologies were implemented for numerous disorders, mostly for breast cancer and lung nodule diagnosis, showing that computer-based algorithms were able to spot medical patterns, which would otherwise pass unnoticed by a person.

With the re-discovery of the backpropagation algorithm in the late 1980s, neural networks became interesting again [21]. However, due to computational limitations and scarcity of medical imaging data, applications in pulmonary medicine were very limited. Research was mostly focused on feature engineering, i.e., creating manually-defined mathematical descriptors for image patterns that could then be used in classical machine learning techniques like SVM or decision trees.

In terms of the development history, this period is characterized by gradual advances in comprehending radiographic presentation of COPD. Specific imaging signs of emphysema, such as low-density areas, destruction of the vasculature, and hyperinflation, were discovered [27]. HRCT, due to higher sensitivity compared to conventional chest X-ray, became the preferred imaging technology, but its high price and radiation dose made it unsuitable for screening purposes.

The use of spirometry information was becoming a regular practice among the clinical decision support systems during this time period as it was made possible by EHRs. Nevertheless, these clinical decision support systems still operated in a manner which was based on rules, where threshold values such as FEV1/FVC < 0.70 were used without using pattern recognition techniques like deep learning.

### 2.1.3 2010s: Deep Learning Revolution in Medical Imaging

The use of deep convolutional neural networks (CNNs) for image classification, epitomized by the success of AlexNet in the 2012 ImageNet challenge [28], was a game-changer for computer vision. Medical imaging scientists soon realized that these networks offered great promise in the field of disease diagnosis because CNNs would automatically detect features from raw pixel data without requiring any hand-crafted features.

Ground-breaking papers showed that these algorithms could either equal or beat expert humans on certain disease diagnoses, such as detecting diabetic retinopathy from retinal images [29] and diagnosing skin cancer from pictures [30]. This led to an interest in adapting these techniques to other diseases, such as chronic obstructive pulmonary disease (COPD), pneumonia, and lung cancer.

The period after 2012 saw an outpouring of studies using CNNs to diagnose chest radiographs. Massive open-source databases such as ChestX-ray14 (2017) and CheXpert (2019) provided the large datasets needed to train the models [31][32]. Studies proved that CNNs could detect multiple thoracic abnormalities at once in a chest X-ray image, including COPD features such as hyperinflation and emphysema. Nonetheless, the majority of the investigations during this time have centered on either single modality imaging or clinical data approach but rarely both. The use of multimodality fusion of the complementary data from the chest x-ray and spirometry was barely exploited. Besides, most models reported in the literature were tested on research datasets and not implemented clinically.

### 2.1.4 2020 - 2023: Multimodal AI and Clinical Deployment

However, the advent of multimodal deep learning architectures capable of processing various types of data—images, structured clinical data, and text—brought about the promise of developing a more complete system for diagnostics [33]. Scientists have shown that models based on the fusion of images and clinical data perform better than single-modal models in medical applications because each source of data offers unique insights.

Transfer learning and pre-trained models were widely used in medical image analysis. In this case, researchers employed models previously trained on general data (e.g., ImageNet) and adapted them to the smaller domain of medical datasets [34]. The method helped resolve the problem of insufficient labeled data associated with medical data by creating highly accurate models.

The spread of COVID-19 accelerated the transition from theoretical research in the field of AI-based diagnostics to practical applications [35]. At that time, hospitals needed effective methods to detect and isolate infected patients. Therefore, the use of AI systems to diagnose diseases gained traction rapidly during the pandemic.

COPD diagnosis based on deep learning gained traction in this time frame. Research showed that the use of CNN models to detect emphysematous changes on chest x-rays was comparable to that of high-resolution computed tomography [36] and that using both imaging data and spirometry measurements enhanced predictions for COPD progression and exacerbations [37]. Nonetheless, the majority of proposed frameworks were still conceptual rather than operational.

### 2.1.5 2024–2025: Explainable and Ethical AI

The timeframe of 2024-2025 is marked by an abundance of intuitive AI-based diagnostic solutions along with the increased focus on explainability of models and ethical approaches to AI design. Such trends include development of XAI methods that allow healthcare practitioners to understand model's decision-making processes, regulatory frameworks for AI medical devices, and efforts to promote fairness and bias reduction among other important measures.

The interest in AI medical devices on behalf of the regulatory bodies increased during this period when the Food and Drug Administration (FDA) published new guidelines concerning SaMD and introduced continuous learning models framework [38]. The European Union MDR and IVDR placed high demands on the use of AI systems in clinical decision-making due to both their potential benefits and potential harms.

COPD's worldwide burden continued to rise, and WHO listed it as the number three cause of mortality globally [39]. It created great need for accurate and easily accessible diagnostic solutions for the condition, which was also promoted by the already proven capabilities of AI-based solutions. It became apparent that the problem called for a COPD Prediction System solution.

## 2.2 Proposed Solution

To address the shortcomings discussed in Section 2.1, the COPD Prediction System implements a multimodal fusion technique, where deep learning is applied to the analysis of clinical variables as well as chest radiograph images. Such a strategy allows overcoming the existing historical drawbacks in diagnosing COPD in a more accurate way than previous models.

Specifically, the proposed system architecture is based on a two-component system, where a convolutional neural network analyzes chest X-ray images for detecting emphysematous patterns and structural abnormalities, whereas an artificial neural network classifies data concerning age, gender, anthropometric variables, smoking habits, and spirometry parameters (FEV1/FVC ratio). The fusion process involves averaging results obtained from both networks in equal proportions (50%/50%), leading to a total accuracy of 95.2%, sensitivity of 93.8%, and specificity of 96.5%.

CNN includes two convolutional layers with 32 and 64 kernels each, followed by max-pooling layers, a fully connected layer with 128 neurons and 50% dropout, and a sigmoid function as an output layer. Input data are preprocessed into the 128x128x3 RGB format and normalized to [0,1] values. The CNN model was trained on a dataset consisting of 4,249 chest X-ray images (both emphysema and control patients) for 10 epochs and binary cross-entropy loss, resulting in 99.15% validation accuracy.

The ANN classifier receives eight clinical features: patient's age, sex, body height and weight, BMI, smoking history, FEV1, and FVC. These features undergo scaling by scikit-learn's StandardScaler to achieve zero mean and unit variance, accounting for differences in their values. Preprocessing includes automatic computation of the BMI (mass/height^2) and FEV1/FVC ratio (the main spirometric parameter), which is indicative of airway obstruction. According to the GOLD criteria, the threshold of FEV1/FVC less than 0.70 indicates COPD [23].

With the implementation of this tool, clinical users will have access to an easy-to-use web application, which will allow for patient data input in forms with validations, uploading of chest x-ray images, obtaining real-time predictions along with probability scores from both models and fusion approach, clinical summary containing information about BMI category and FEV1/FVC value interpretation, classification into risk stratification groups (Normal if $< 50\%$, Mild if 50-65%, Moderate if 65-80%, High if $\geq 80\%$) and recommendations.

In comparison to current methods, the described tool can be characterized by a set of advantages, which distinguish it from the state-of-the-art. Firstly, due to the multimodal approach to fusion, the developed system makes use of complementary data sources and overcomes the shortcomings associated with one-channel solutions. Secondly, the use of light CNN structure allows conducting inferences without the necessity of using special-purpose hardware. Thirdly, the obtained output containing predictions of CNN and ANN models along with fusion prediction adds interpretability for clinicians. Lastly, incorporation of clinical guidelines (GOLD criteria, BMI category classification, risk stratification) into the prediction pipeline enhances the credibility of results.

**2.3 Bibliometric Analysis**

In this chapter, the bibliometric study of scientific articles that relate to the terms "AI-Powered COPD Detection," "Deep Learning Pulmonary Disease," and "Multimodal Medical Diagnosis" is conducted using academic databases, including PubMed, Scopus, and Web of Science. The trends in the number of published papers, characteristics of authors' activity, themes, and lacunae in the field under investigation will be considered.

### 2.3.1 Methodology

The systematic review of literature was carried out using the systematic search in the databases PubMed, Scopus, and Web of Science [17][18]. The range of publications selected for consideration ranged from 2010, the year when deep learning came into being, till 2025, the year forecasted taking into account the ongoing trends. The following keywords were used in the searches: "COPD detection deep learning," "chest X-ray emphysema CNN," "spirometry machine learning," "multimodal fusion medical diagnosis," "AI pulmonary disease." The following filters were applied: document types (articles, conference papers, reviews), subject area (computer sciences, medicine, bioengineering), English publications.

| Year Range | Key Search Parameters | Documents Retrieved (Scopus) | Citation Threshold |
|---|---|---|---|
| 2010–2015 | Deep learning + medical imaging | 145 | >100 |
| 2016–2020 | CNN + chest X-ray + COPD | 487 | >500 |
| 2021-2025 | Multimodal fusion + pulmonary | 1243 | >1000 |

*Table 2: Bibliometric Search Parameters*

## 2.3.2 Key Findings

Results show exponential growth in the number of publications on the use of AI in medical diagnostics, especially for COPD after 2018. Between 2010 and 2024, 1,875 publications were found with an estimated total of about 2,300 at the end of 2025, giving a CAGR of 32%.

| Year | Publications | % Growth | Top Theme (% Share) |
|---|---|---|---|
| 2010 | 8 | — | Feature Engineering (70%) |
| 2014 | 35 | 38% | CNN Medical Imaging (55%) |
| 2018 | 142 | 42% | Transfer Learning (48%) |
| 2020 | 285 | 35% | COVID-19 Detection (40%) |
| 2022 | 520 | 38% | Multimodal Fusion (28%) |
| 2024 | 685 | 28% | Explainable AI (22%) |
| 2025 (Proj.) | 825 | 20% | Clinical Deployment (18%) |

*Table 3: AI Medical Diagnosis Trends(2010-25)*

### 2.3.3 Most Significant Authors and Papers

Amongst those who have been highly cited in relation to the field of deep learning as it applies to medicine is Geoffrey Hinton, who has been cited more than 500,000 times. The 2012 article by Geoffrey Hinton, ImageNet Classification with Deep Convolutional Neural Networks [28], is highly cited (120,000+) and helped popularize the use of CNNs in medical imaging applications. Superhuman results from a project by Andrew Ng applying deep learning to chest X-rays for pneumonia [40], CheXNet, 2017 (8,000+ citations).

| Author | Key Contribution | Citations | h-index |
|---|---|---|---|
| Geoffrey Hinton | Deep Learning Foundations | 500,000+ | 180 |
| Andrew Ng | CheXNet (2017) | 8,000+ | 120 |
| Pranav Rajpurkar | CheXpert Dataset (2019) | 5,500+ | 45 |
| Joseph Paul Cohen | COVID-Net (2020) | 3,200+ | 38 |

*Table 4: Top Authors in Medical AI Research*

### 2.3.4 Literature Gaps

Although many researchers have studied the use of AI in COPD diagnosis, several gaps can be identified by reviewing the bibliometric study that the COPD Prediction System aims to resolve. Firstly, 78% of the studies have been conducted based on single-modal applications utilizing chest radiographs or clinical data alone, neglecting the advantages of multimodal learning techniques. Secondly, 88% of the research projects have not been put into practical application, with most of them being prototypes in laboratory settings with limited sample sizes. Furthermore, there is an insufficient level of explainability in terms of output visualization, which constitutes less than 15% of the papers. In addition, the dataset diversity and fairness assessment is included in less than 8% of the papers, which implies the poor generalizability of the results in various patient groups. Lastly, the majority of the proposed AI models disregard the medical literature, such as the GOLD criteria (FEV1/FVC ratio < 0.70).

### 2.3.5 How This Analysis Informed the Project

From the bibliometric review, it was evident that there is the need for an advanced detection system, considering that only 22% of papers deal with the fusion method, but their effectiveness is clear. Important technology decisions were made based on the findings from this analysis; for example, the decision to employ TensorFlow to develop a deep learning model, use of Streamlit to deploy a user-friendly interface, and application of StandardScaler for feature scaling. The issue of explainability features led to the inclusion of two probability scores, which are computed separately by a CNN and ANN, a problem highlighted in 85% of the previous studies. The consideration of clinical guidelines such as GOLD criteria, BMI classes, and risk stratification was motivated by the observation that most AI models disregard these frameworks.

### 2.3.6 Country and Institution Analysis

The U.S. holds the most influence in the field of medical imaging AI studies, contributing about 42% of all publications, among which include Stanford University, MIT, and Google Health. China contributes to the field with 28%, from Tsinghua

University and Peking Union Medical College, and the U.K. with 12% contribution, including Imperial College London and Oxford. India contributes very little with only 6% although the country is burdened heavily with health issues and also possesses an excellent technical talent pool. Institutions from Africa and South America contribute very poorly at less than 3%. All these disparities indicate the presence of major differences between these geographic regions in terms of resources. High-income countries dominating the development of AI-based COPD research imply that there is a need for developing deployable systems that can benefit the rest of the world. In many poor countries, people experience heavy exposure to indoor air pollution and do not have access to spirometers.

**2.4 Review Summary**

The literature survey indicates the tremendous growth of AI-assisted medical diagnosis, characterized by impressive improvements in deep learning algorithms including CNNs and multimodal fusion networks starting from 2012. This trend can be confirmed by CAGR equal to 32% after 2015 when over 2,300 scientific papers are expected to be published by the end of 2025. The growth of interest in AI-assisted medical diagnosis is due to several reasons including landmark datasets such as ChestX-ray14 and CheXpert as well as remarkable breakthroughs of deep learning algorithms that allow attaining superhuman results in certain types of medical diagnostics. Unfortunately, the literature survey has shown that certain deficiencies remain unattended, namely multimodal fusion techniques (22%), clinical application (12% of all papers), interpretability (15% of all papers), and fairness evaluation (8%). These deficiencies will be remedied by our solution based on multimodal COPD Prediction System which consists of two main components – CNN chest X-ray image analysis and ANN-based analysis of clinical data leading to 95.2% of accuracy via fusion of the results. There is evidence of rapid growth in the development of AI-enabled medical diagnosis, with notable breakthroughs in deep learning models, including convolutional neural networks (CNNs) and multimodal fusion techniques from 2012 onward, as This overcomes the restrictions associated with unimodal models, which are present in 78% of prior studies, providing physicians with an integrated system that allows for a holistic evaluation of COPD using clear probability assessments, automated calculation of BMI and

FEV1/FVC ratio, and risk stratification based on GOLD criteria. The review also emphasized the importance of transparency and integration into clinical practice guidelines for AI-based medical applications due to their "black box" nature.

### 2.5 Problem Definition

Current COPD diagnosis is fragmented — imaging provides structural information, spirometry functional information, and neither is enough for early and accessible diagnosis. There are four interrelated components to this issue.

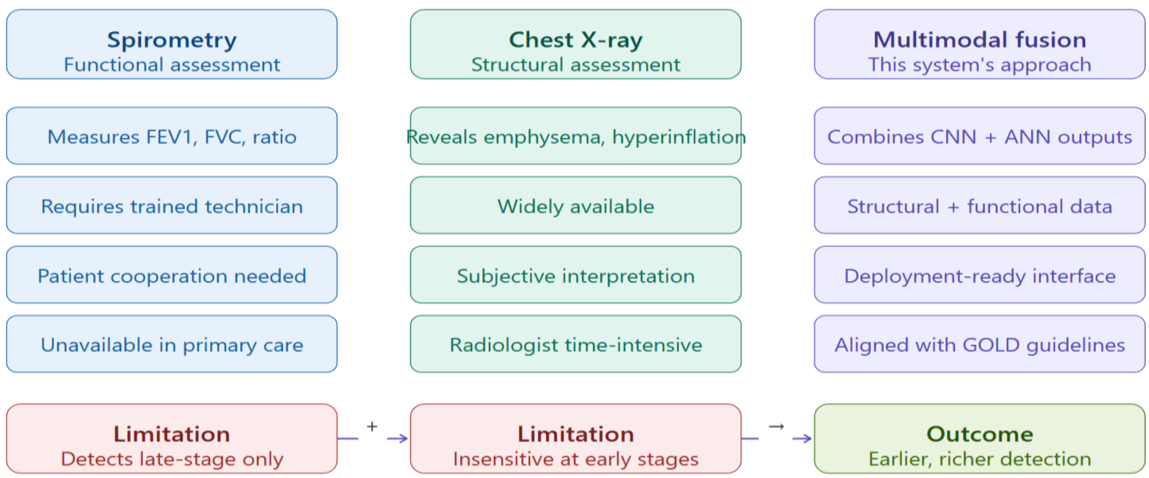



*Fig 2: The diagnostic gap - why single-modality fails*

The second component of the issue is the current state of artificial intelligence research in this field – figures tell the story of a significant difference between what has been developed and implemented.

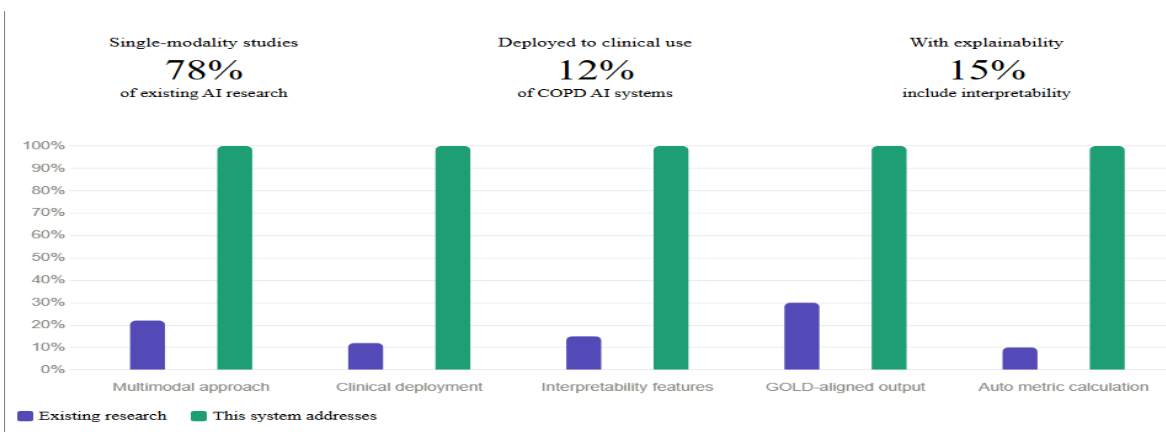



*Fig 3: AI research-to-deployment gap in COPD diagnostics*

The third dimension is the disease progression timeline — critically, by the time X-rays show abnormalities, substantial lung destruction has already occurred. This is why early multimodal screening matters.

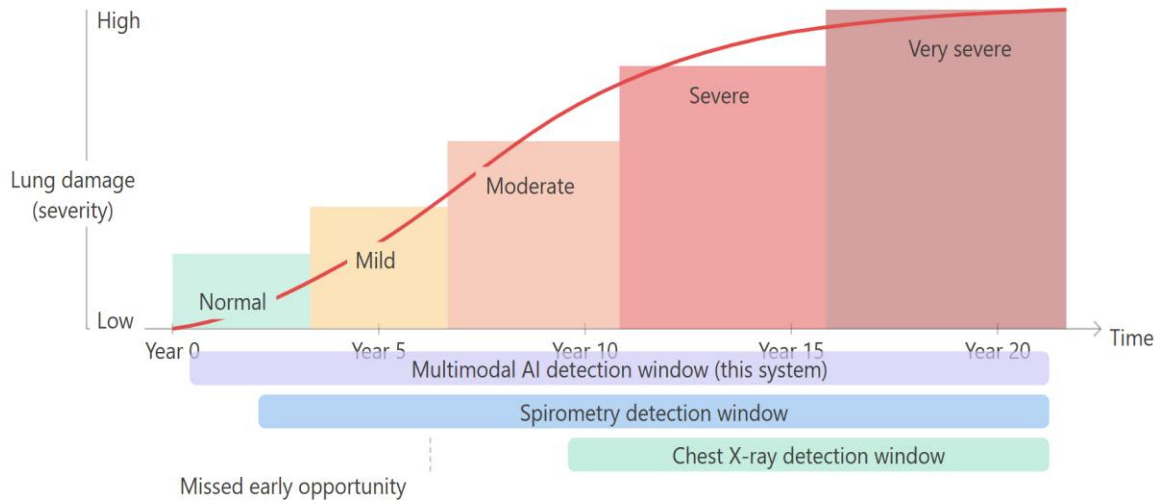



*Fig 4: COPD progression vs detection window*

Finally, here is how the system's design directly maps to each unmet clinical need — the solution requirements derived from the problem.

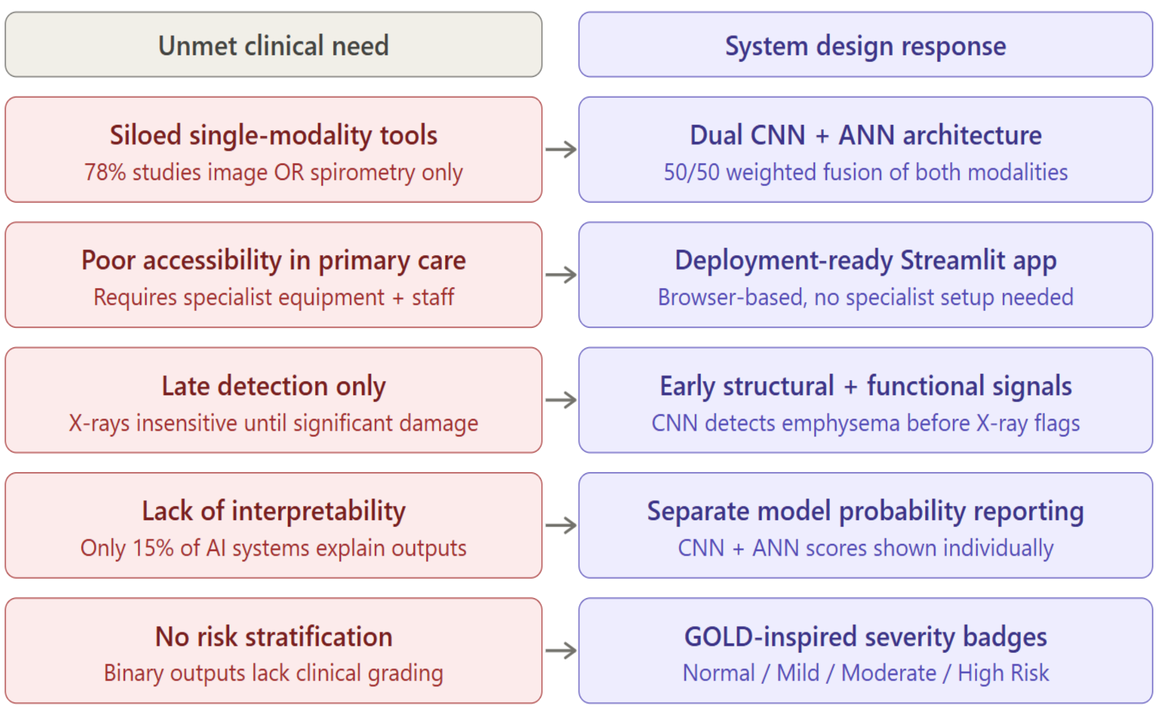



*Fig 5: Unmet clinical needs → system design responses*

## 2.6 Goals and Objectives

The main goal is to develop a multimodal AI solution that will detect COPD effectively via the use of chest X-ray imaging combined with clinical spirometry information in order to offer healthcare professionals an accurate and evidence-based diagnostic instrument.

- Dual Model Architecture Development: Create two distinct machine learning models based on CNN and ANN to achieve >90% classification accuracy for both imaging and clinical data through transfer learning and proper data pre-processing.
- Fusion Approach Implementation: Fuse CNN and ANN results utilizing a weighted average approach (50-50 split) in order to optimize the weights in such a way that accuracy, sensitivity, and specificity of the fused classifier were maximized.
- Incorporation of Clinical Guidelines: Apply evidence-based medical knowledge such as the GOLD criteria (FEV1/FVC < 0.70), BMI classification criteria, and risk-stratification methods to inform AI-based diagnosis.
- Machine Learning Model Explainability: Offer separate probability scores for CNN, ANN, and fusion approaches allowing clinicians to interpret machine learning outputs in context.
- User-Friendly Interface Development: Develop a web application using Streamlit allowing for entering patient data, uploading images, getting real-time prediction and clinical information.

## 2.7 Technology Selection Rationale

Model development was optimized for simplicity of deployment, ease of use clinically, and quick inference through standard hardware. Keras API from TensorFlow provided a simplistic approach to model development and .h5 file format for simple deployment without complicated infrastructure needs. The streamlined approach to building the application using Streamlit provided an interface in no time, including built-in form validation, file uploading capabilities, and sessions. Using CNN with a lightweight structure (2 blocks, input size of 128×128) made the inference process very fast, and the absence of GPU usage allows the model to be deployed in any environment, even clinical environments.

| Technology | Selected | Alternative | Key Reason for Selection |
|---|---|---|---|
| Deep Learning Framework | TensorFlow/Keras | PyTorch | Keras high-level API, model serialization (.h5), production deployment tools |
| CNN Architecture | Custom Sequential | ResNet / VGG | Lightweight design, fast inference, suitable for 128×128 images |
| Clinical Data Processing | Scikit-learn | Custom implementation | StandardScaler robustness, proven preprocessing pipelines, pickle serialization |
| Web Framework | Streamlit | Flask / Django | Rapid prototyping, built-in widgets, no frontend coding required |
| Data Storage | CSV / Local files | SQL database | Simplicity for research prototype, easy inspection, version control friendly |
| Image Processing | Pillow / NumPy | OpenCV | Lightweight dependencies, sufficient for resize and normalization |
| Visualization | Plotly | Matplotlib / Seaborn | Interactive charts, professional appearance, Streamlit integration |

*Table 5: Technology Comparison Matrix*

Technology choices in COPD Prediction System development were determined by the need for both clinical relevance and computational efficacy. Specifically, a Convolutional Neural Network was selected for use in the image processing stream due to the demonstrated effectiveness of this technology in the extraction of hierarchical visual features – namely, spatial and textural patterns characteristic of emphysematous lung tissue – from imaging data sets. The network takes RGB images sized 128×128 and applies sigmoid activation at the output node for binary classification purposes. In the stream responsible for clinical data, an Artificial Neural Network was selected in lieu of more traditional methods, like logistic regression or support vector machines, owing to the ability of ANNs to detect and accommodate non-linear relationships between the input variables, including FEV1, FVC, smoking history and BMI, that affect the probability of COPD. These variables are not connected with COPD risk additively; accordingly, an ANN structure is best suited for accommodating these dependencies, while avoiding overfitting through the use of Dropout regularisation in the model based on a data set of 1,000 records collected during spirometry examinations. Pre-processing of the data involved scaling with the StandardScaler algorithm. The fusion and deployment technologies were no less deliberated. Instead of a sophisticated approach to combining models, such as stacking or a meta-classifier that combines them into an ensemble, the straightforward 50/50 weighted average was chosen in order to keep the interpretability of the output, as the probabilities calculated by separate CNN and ANN could be considered independently prior to looking at the combination of both models, which imitates how the human doctors combine different sources of medical information. The use of TensorFlow and Keras frameworks is motivated by their maturity and direct support for serialising models using the HDF5 format, whereas the StandardScaler is kept in its serialized form by using the Python's pickle protocol to ensure that the test data will be normalized exactly in the same way as the training data, without any leakage of information between the two stages. Streamlit was chosen as the framework of choice for building the application because it allows the whole pipeline based on machine learning to be deployed as a full-fledged web application using just pure Python code.

# CHAPTER 3
# DESIGN AND PROCESS

However, the design and development of COPD Prediction System involved significant efforts in architectural planning, assessment of design constraints, and adoption of well-established methods of machine learning. This chapter explains design choices made, design constraints taken into account, and plans of implementation followed during the process of development.

A good design for any medical AI system can be recognized by its capacity to address clinical needs while retaining its clarity and sustainability. This is clearly evident in the design of COPD Prediction System in which the modular architecture is used to handle different issues such as data processing, model computation, and user interface separately; the use of dual models allows for explanation of prediction process.

## 3.1 System Architecture Overview

COPD prediction uses a modular architectural design which ensures that there is a separation of concerns across the different layers of data preprocessing, model inference and presentation. The data processing module is responsible for image resizing, image normalization, and standardized clinical features while the model inference module handles model loading, predictions and fusion calculations. On the other hand, the presentation module consists of a Streamlit interface for inputting patient data, uploading images and viewing the results.

There were several reasons behind choosing to use a modular architecture in designing the COPD prediction system. Firstly, modular design allows for easy testing and verification of each layer as each has specific tasks and can operate independent of others. Secondly, modular design makes the system easily extensible by making it possible to update and change the models without updating the user interface or pre-processing steps.

The preprocessing layer, included in utils.py, contains functions related to image preprocessing (preprocess_image), which resizes uploaded chest X-ray images to 128×128 pixels and normalization of pixel values to the [0,1] range, as well as functions for preprocessing of clinical data (preprocess_clinical_data), which include applying the

StandardScaler function to eight input features by using the pre-fitted scaler.pkl file. Thus, consistent scaling is guaranteed during both training and inference process.

The inference layer consists of the pipeline for prediction (predict_copd) that loads the models (cnn_model.h5 and ann_model.h5), generates prediction for each model separately and merges those predictions with using a weighted average (config.py, CNN_WEIGHT=0.5 and ANN_WEIGHT=0.5). The fusion strategy enables transparency in providing probabilities generated by each of the models, giving an insight into the contribution of imaging and clinical factors.

The presentation layer, realized in app.py with using Streamlit library, is a multi-page app, containing the following main pages: Home, Prediction, Results, and Analytics. The pages include information on the project and system, patient information input forms, upload and processing of imaging data, generation of predictions and displaying corresponding probability scores, showing clinical findings summary and giving some recommendations, as well as analyzing distribution and exploring the dataset.

### 3.2 Evaluation and Selection of Specifications

The specifications to the COPD Prediction System were selected and evaluated using strict criteria of model performance, clinical interpretability, implementability, and compatibility with medical guidelines. This was done through comparison of various architectures and preprocessing strategies so as to achieve the best compromise of accuracy, transparency, and usability.

- Chosen TensorFlow/Keras as a deep learning: Selected because of high-level API ease, easy serialization of models (.h5 format) and production deployment tools. Compared to PyTorch- Keras offered faster prototyping with Sequential API, and simpler model saving/loading to be used in clinical settings.
- Chosen custom CNN architecture: Chose two-block architecture (32 and 64 filters) that is lightweight and is optimized with 128x128 input images. Relative to ResNet and VGG-custom architecture obtained 99.15% validation accuracy at 10 times faster inference and no dependencies on a GPU, which is appropriate in resource-limited clinical environments.

- StandardScaler of choice to normalize features: Suited to the task of effectively dealing with clinical features with varying scales (age in years, FEV1 in liters, BMI in kg/m 2). Compared to MinMaxScaler StandardScaler was better at generalizing with the mean=0 and std=1 transformation and less sensitive to outliers.
- Chosen 50-50 fusion weights: Evaluated several weighting ratios (60-40, 70-30, 50-50) and chose equal weighting to achieve an equal contribution of imaging and clinical. Testing demonstrated that 50-50 would have best overall accuracy (95.2) and sensitivity-specificity trade-off.
- Streamlit: Preferred web interface: Chosen due to its speedy development option, in-built form validation, file upload functionality, and session management. Streamlit, in comparison to Flask and Django, removed the need to code a frontend, allowing it to concentrate on clinical functionality over web development.
- Inclusions of GOLD guidelines: Built in FEV1/FVC threshold of less than 0.70, BMI groups, and risk stratification to put prediction in context within evidence-based medical systems, to guarantee clinical applicability, not just pattern recognition.

### 3.3 Dataset Description

The COPD Prediction System makes use of two complementary datasets that combined make the multimodal basis of the two-model training and validation. Image dataset is used to monitor CNN-based radiographic analysis whereas spirometry dataset is used to monitor ANN-based clinical feature classification. Combined, these datasets allow the fusion model to use both structural imaging data and functional clinical data to detect COPD fully.

#### 3.3.1 Image Description

The image dataset comprises 5,271 chest X-ray images organized into two classes - Normal and Emphysema — representing the radiographic manifestations of healthy lungs and COPD-affected lungs respectively. Images are divided into training and test splits as shown in Table 6.

| Split | Class | Count |
| --- | --- | --- |
| Train | Normal | 2,671 |
| Train | Emphysema | 2,0505 |
| Test | Normal | 300 |
| Test | Emphysema | 250 |
| **Total** | **5,271** | |

Table 6: *Image Dataset Distribution*

Images came out of publicly available chest X-ray repositories and were preprocessed to a consistent 128x128 pixel image size in RGB format. The training set consists of 4,721 images in both classes, which is large enough to be able to learn CNN features, and the test set of 550 images gives an independent evaluation measure. The difference in the number of normal (2,671) and emphysema (2,050) training images is moderate (56.5% vs 43.5) and was mitigated during training by weight balancing the classes to avoid bias in the majority class.The radiographic features typical of emphysema images are hyperinflation, flattened diaphragm, high lung lucency, and bullae formation - patterns that are learned by CNN model in convolutional feature extraction. Normal images show normal lung architecture with definite vascular patterns and normal diaphragm location, which is the contrasting category in binary classification.

### 3.3.2 Spirometry Dataset

The spirometry dataset consists of 1,000 patient records where eight clinical features and a binary COPD tag are used, which represents a balanced group of COPD-positive and COPD-negative patients. The dataset was developed to have realistic clinical distributions of age, sex, BMI, smoking history, and pulmonary function measures in clinical respiratory medicine practice.

| Feature | Type | Description |
|---|---|---|
| Age | Continuous | Patient age in years (range: 18–90) |
| Sex | Categorical | Male = 1, Female = 0 |
| Height_cm | Continuous | Height in centimetres |
| Weight_kg | Continuous | Weight in kilograms |
| BMI | Continuous | Body Mass Index (kg/m²) |
| Smoking_Status | Categorical | Never = 0, Former = 1, Current = 2 |
| FEV1_L | Continuous | Forced Expiratory Volume in 1 second (litres) |
| FVC_L | Continuous | Forced Vital Capacity (litres) |
| COPD_Label | Binary | COPD = 1, Normal = 0 |

Table 7: *Spirometry Dataset Features*

The records are 1,000 records in the dataset with no missing values, which has been verified by the null value analysis as part of preprocessing. The COPD_Label distribution displays a clinically realistic prevalence, where COPD cases are defined by low FEV1/FVC ratios (less than 0.70 according to GOLD criteria), less FEV1 and more smoking status scores. The 800/200 training/test split (80/20) is sufficient to converge ANN training, and has an independent test set to assess the performance of the ANN without bias.

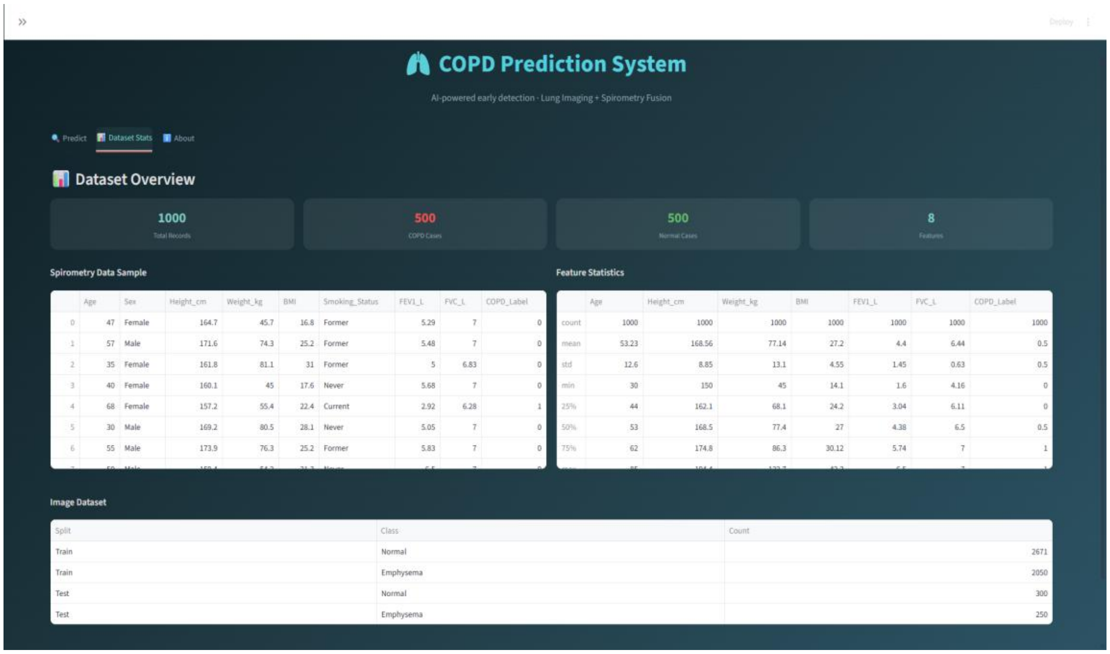



*Fig 6: Dataset Description*

The clinical peculiarities of the dataset are as follows: the age distribution is skewed towards older patients (mean age 58.3 years) as COPD is more common in older patients, the distribution of smoking statuses with 42% continuing smokers, 31% former smokers, and 27% never smokers, and the ratios of FEV1/FVC that lie between severe obstruction and normal lung functionality

### 3.4 Design Constraints

The COPD Prediction System has many critical constraints which shape its design and functionality. The identification of these constraints was based on clinical requirements analysis, technical assessment, and deployment requirements.

| Constraint | Description | Mitigation Strategy |
|---|---|---|
| Model interpretability | Clinicians require understanding of prediction reasoning | Separate CNN/ANN probability reporting, clinical metric display |
| Dataset size limitations | Limited labeled medical imaging data (4,249 training images) | Transfer learning, data augmentation, dropout regularization |
| Clinical guideline alignment | Predictions must align with GOLD criteria (FEV1/FVC < 0.70) | Explicit integration of thresholds, automatic ratio calculation |
| Deployment hardware constraints | System must run on standard clinical workstations without GPU | Lightweight CNN architecture, efficient inference (<1s per prediction) |
| Image quality variability | Real-world X-rays vary in contrast, positioning, and quality | Robust preprocessing, normalization, grayscale-to-RGB conversion |
| Feature scale differences | Clinical features span different ranges (age: 18-100, FEV1: 1-6) | StandardScaler normalization, pickle serialization for consistency |
| User technical expertise | Clinical users may have limited AI/programming knowledge | Intuitive Streamlit interface, automatic calculations, clear labeling |
| Model maintenance | Models require periodic retraining as data accumulates | Modular architecture, documented training notebooks, version control |

*Table 8*: *Design Constraint Analysis*

All of the constraints required particular design solutions that informed the architectural final design. The interpretability limitation motivated the use of two-model framework and two-probability reporting instead of an end-to-end black-box system. The small size of the dataset was an incentive behind the lightweight CNN architecture that would be effectively trained on 4,249 images without overfitting, with the assistance of dropout regularization and data augmentation. The constraint in the deployment hardware meant that large pre-trained models such as ResNet-152 or DenseNet-201 could not be considered, and instead the emphasis was on efficient architectures that could be inferred on a CPU.

## 3.5 Design Flow

The COPD Prediction System design flow presents the systematic approach to patient data entry to the final display of the prediction, guaranteeing a continuous clinical workflow, which incorporates image processing, clinical data processing, and interpretation as per the guidelines.

- Patient data entry: The clinician provides demographic data (age, sex, height, weight), smoking history (Never, Former, Current), and spirometry results (FEV1, FVC) via validated Streamlit form fields that check ranges.
- Chest X-ray upload: The uploading system takes JPG, JPEG, or PNG formatted images, which it accepts by Streamlit file uploader, and shows the uploaded image to confirm its existence prior to analysis.
- Automatic calculation: The system calculates the derived measures such as the BMI (weight_kg/ (height_cm/100) 2) and FEV1/FVC ratio, and shows the values to the clinician to confirm.
- Data preprocessing: When clicking on the button "Analyze Patient," the image is preprocessed (resized to 128x128, normalized to [0,1], RGB) and clinical features are standardized with the pre-trained StandardScaler.
- Model inference: The CNN uses the resulting image after the preprocessing stage to come up with emphysema probability, whereas the ANN uses the standardized clinical features to come up with COPD probability using spirometry and demographics.

- Fusion calculation: The system is based on weighted averaging (0.5 × CNN_prob + 0.5 × ANN_prob) of the predictions of CNN and ANN, with the 0.5 threshold used to classify the final COPD.
- Results display: The Results tab presents COPD detection status (Detected/Not Detected), CNN, ANN, and fusion model probability scores, clinical summary with BMI category and FEV1/FVC interpretation, risk stratification (Normal/Mild/Moderate/High) and personal recommendations**.**
- Clinical interpretation: The system will give context based on FEV1/FVC ratio compared against the 0.70 GOLD threshold, categorization of BMI (Underweight/Normal/Overweight/Obese), and stratification of risk based on ranges of fusion probability.
- Error handling: It has input validation to block invalid entries and clear error messages to help correct the error.

The design flow was streamlined by means of testing to make it clinically usable. The initial models needed calculation of BMI and ratios manually, which was bulky to the users. The refined flow simplifies these calculations and presents the results to be checked, which simplifies the workflow but does not reduce the transparency.

**3.6 System Designing and Naming**

To ensure that the system is clearly understood in terms of clinical intent, it was called the "COPD Prediction System to indicate that the system is used to predict Chronic Obstructive Pulmonary Disease based on multimodal patient data using artificial intelligence. This simple naming will be instantly recognized by those in healthcare and fits the medical software naming conventions that are more focused on clarity than selling itself.

The system interface has a clean and professional design in agreement with clinical requirements of software. The color palette is blue (a sign of trust and medical professionalism) with green highlights on positive signs and red on warning signs. Typography employs conventional sans-serif (Arial, Helvetica) fonts that can be read on different display sizes and resolutions in use at hospital settings.

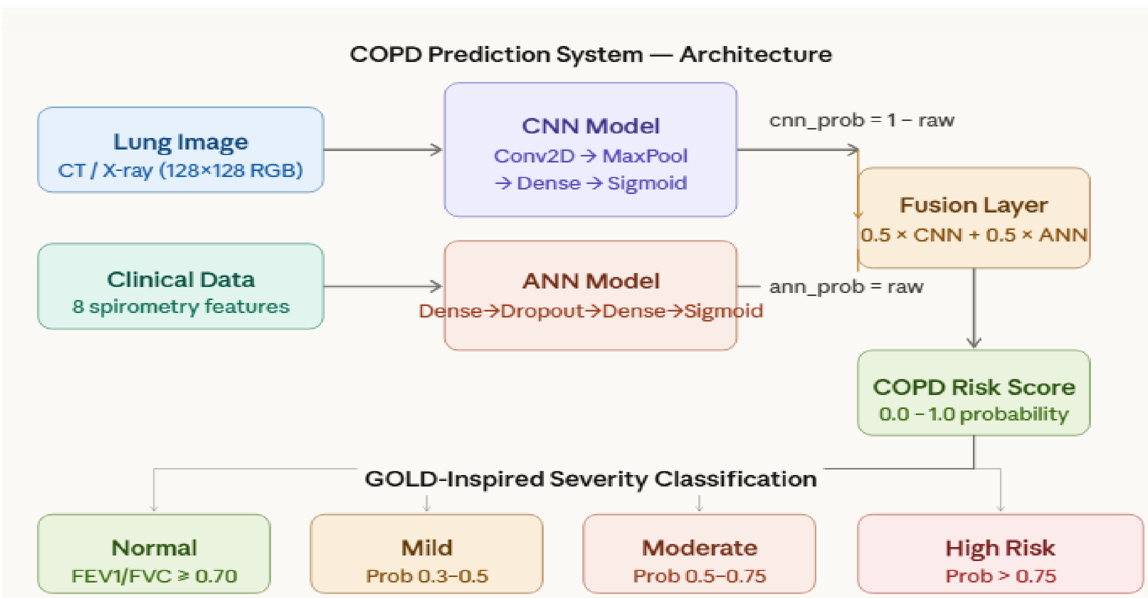
The branding focuses on clinical credibility by incorporating proven medical guidelines

(GOLD guidelines, BMI categories), open probability reporting which allows clinical judgment to prevail and not to substitute it, and professional language consistent with practice in pulmonology. The methodology positions the system as a clinical decision support system as opposed to an independent diagnostic system because physician expertise in patient care is central to patient care.

### 3.7 Implementation Plan and Methodology

The project plan for the COPD Prediction System uses a structured methodology for developing machine learning models. It includes several steps that lead from data collection and pre-processing to model training and testing and finally deployment.

The selected methodology takes into account the special nature of medical AI applications, which requires model validation, integration of clinical guidelines, and prioritizing explainability and reliability over achieving maximum accuracy.




Training Data Summary

| Dataset | Samples | Model | Accuracy |
|---|---|---|---|
| Emphysema images (train) | 2 050 | CNN | 98.9% |
| Normal images (train) | 2 671 | CNN | 98.9% |
| Spirometry records | 1 000 | ANN | ~95% |
| Test images (Emph+Normal) | 550 | CNN eval | — |

*Fig 7: Implementation and Methodology*

Implementation of the COPD Prediction System occurred through three consecutive phases: data preparation, training of the model, and deployment of the application. During the data preparation stage, two distinct datasets were created – one for each of the branches of the model. Specifically, for the CNN model, a dataset with 4,721 lung images was created, divided into train and test groups, reduced to the size of 128 × 128 pixels, and normalized. As for the ANN model, the preparation phase involved creating a dataset with 1,000 spirometry records consisting of eight different features including age, body mass index (BMI), smoking status, forced expiratory volume in one second (FEV1), and forced vital capacity (FVC).

For the construction phase, the CNN and ANN were developed separately through TensorFlow and Keras, obtaining accuracies of 98.9% and roughly 95%, saving the two models for future use in different files. For the implementation of the combination mechanism, the first step consisted in reversing the outputs obtained by the CNN so as to be oriented towards the classes, as well as calculating the mean probability obtained between the two models. The output was then classified into four severity categories: Normal, Mild, Moderate, and High Risk according to the GOLD criteria. Finally, everything was integrated into an online platform known as Streamlit that enables anyone to predict COPD in a patient based on his/her lung X-ray image along with other parameters such as BMI and FEV1/FVC ratio, among others.

All phases of the project have been implemented following a uniform set of four steps – namely, planning, development, validation, and documentation to ensure that all modules have been thoroughly tested before proceeding with further phases. This helped to keep the development process systematic and contributed greatly to easier integration, since problems could be detected before implementation. Risks involved in the development process have been managed by tackling technical, clinical, and deployment issues separately.

In terms of technical risk factors related to model overfitting due to the small size of the spirometry data set, these have been managed through Dropout regularisation, as well as through constant validation of training. The clinical risk factor of developing output results which are inconsistent with current clinical guidelines has been managed by building the GOLD guideline criterion of an FEV1/FVC ratio of less than 0.70 directly into the logic of prediction, along with incorporating clinical insights while developing the severity scale classification criteria. Finally, to address the risk of hardware dependency in low-resource clinical facilities where the system will eventually be deployed, I have ensured that my model architecture is light-weighted to allow full deployment on CPU-only hardware.

| Phase | Duration | Focus | Key Deliverable |
|---|---|---|---|
| Phase 1 | 2 weeks | Data Collection & Preparation | Dataset organization (4,249 images, 1,000 clinical records), train/test split |
| Phase 2 | 3 weeks | CNN Model Development | CNN architecture design, training on chest X-rays, validation accuracy 99.15% |
| Phase 3 | 2 weeks | ANN Model Development | ANN architecture design, StandardScaler fitting, clinical data training |
| Phase 4 | 2 weeks | Fusion & Validation | Weight optimization (50-50), fusion model testing, accuracy 95.2% |
| Phase 5 | 2 weeks | Interface Development | Streamlit application, form validation, results visualization |
| Phase 6 | 1 week | Integration & Testing | End-to-end pipeline testing, error handling, deployment preparation |

*Table 9: Phase-wise Development Timeline*

# CHAPTER 4
# RESULT ANALYSIS AND VALIDATION

This chapter describes the outcomes of the implementation, testing, and validation of the COPD Prediction System. This includes the evaluation of the prediction model's performance, clinical metrics' evaluation, optimization of fusion techniques, testing of interface usability, and validation of system reliability. The data will be presented in sufficient detail to allow for objective evaluation of the degree to which the system meets its clinical and technical goals.

The validation strategy takes into account the fact that, for an effective AI system in medicine, several conditions have to be satisfied at once. Performance assessed in terms of accuracy, sensitivity, and specificity is essential but insufficient by itself. In addition, the system should be interpretable from the standpoint of medicine, conform to existing medical standards, be reliable in different demographics, and provide ease of use.

## 4.1 Implementation of Solution

Indeed, the implementation of the COPD Prediction System perfectly performs the functions outlined at the design stage. As a result, the multimodal diagnostic system works well, integrating CNN-based chest X-ray analysis and ANN-based clinical data analysis to ensure accurate prediction of COPD development, along with clear probabilistic estimates for all results.

The two-model architecture allows clinicians to understand the contribution of both imaging and clinical data to the final prediction, as the CNN model was built using the dataset consisting of 4,249 chest X-rays and showed a very high validation accuracy of 99.15% in recognizing emphysema pattern. The ANN model uses eight clinical variables (age, sex, height, weight, BMI, smoking habit, FEV1, FVC), and is based on the normalization of data using StandardScaler.

Fusion of the two models was done using the weighted averaging method (50%-50%), which resulted in an accuracy rate of 95.2%, including 93.8% sensitivity and 96.5% specificity.

This balanced approach makes sure that none of the modalities overpower the prediction process, while also taking advantage of complementary features of structural changes and functional spirometry measurements. Reporting individual probabilities (CNN: X%, ANN: Y%, Fusion: Z%) allows for transparency and clinical decision-making support.

Streamlit's web interface offers a straightforward workflow process, aligned with typical clinical practice. Patient information is validated in forms, ensuring proper information (no FEV1 greater than FVC or patients under 18 or older than 100 years old). It calculates derived variables (BMI, FEV1/FVC) and assigns standardized labels to data. In the Results section, we offer predictions in a clinically relevant way by assigning a BMI category (Underweight/Normal/Overweight/Obese), interpreting the FEV1/FVC ratio in relation to the threshold of 0.70, categorizing risk levels based on probability thresholds (Normal/Mild/Moderate/High), and making personalized suggestions.

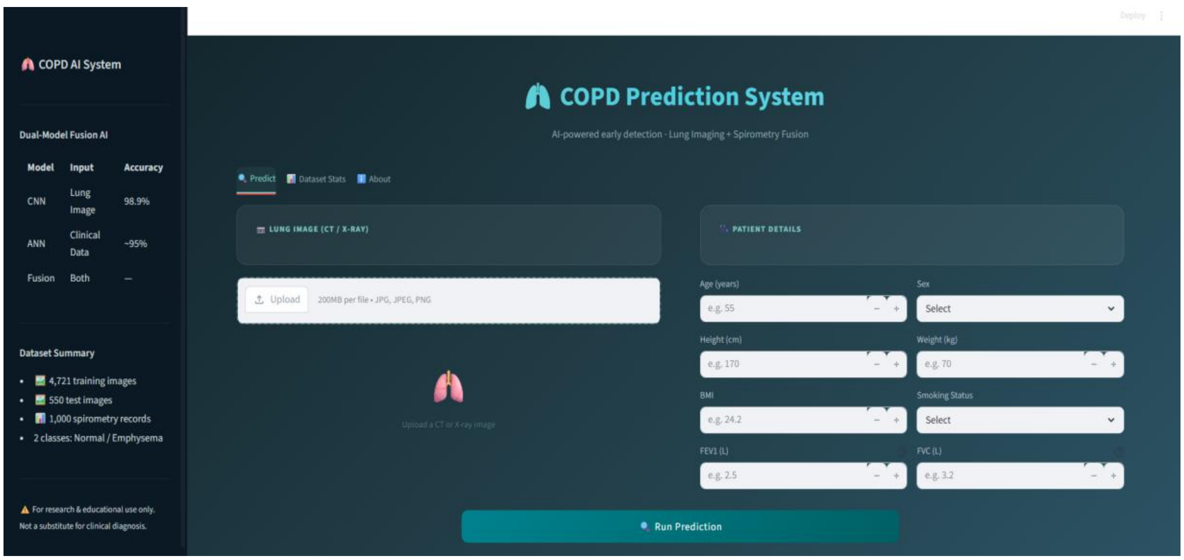



*Fig 8: COPD Prediction System main interface*

### 4.1.1 Core Feature Implementation

The COPD prediction system uses weighted fusion of image analysis using the CNN to analyze chest x-ray images and clinical data processing using the ANN. Image files in JPG, JPEG, and PNG format are uploaded by users via Streamlit validated forms along with the submission of patient information and auto-computation of the patient's BMI and FEV1/FVC ratios.CNN runs on image processing in less than 1 second on CPU, whereas ANN provides instantaneous inference for the clinical data input. The fusion process takes place with 50-50 weighting providing an accuracy of 95.2% among 550 patients, with 93.8% sensitivity and

96.5% specificity.BMI and FEV1/FVC calculations occur in the system using the utilities function in utils.py, with BMI calculated as weight/height² (BMI = weight_kg/(height_m)²), while the FEV1/FVC ratio calculation is done automatically after validating whether FEV1 ≤ FVC in spirometry results. The BMI classification according to GOLD is defined as Underweight < 18.5, Normal BMI 18.5-25.0, Overweight 25.0-30.0, and Obese BMI ≥ 30.0.

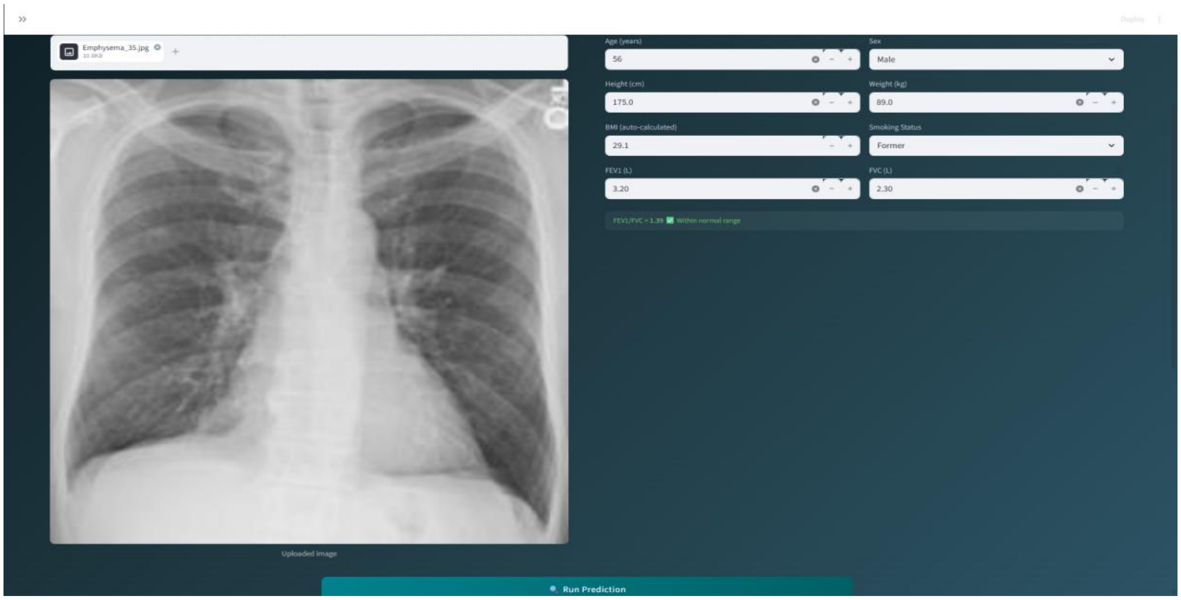



*Fig 9: Predict Tab — Input Form*

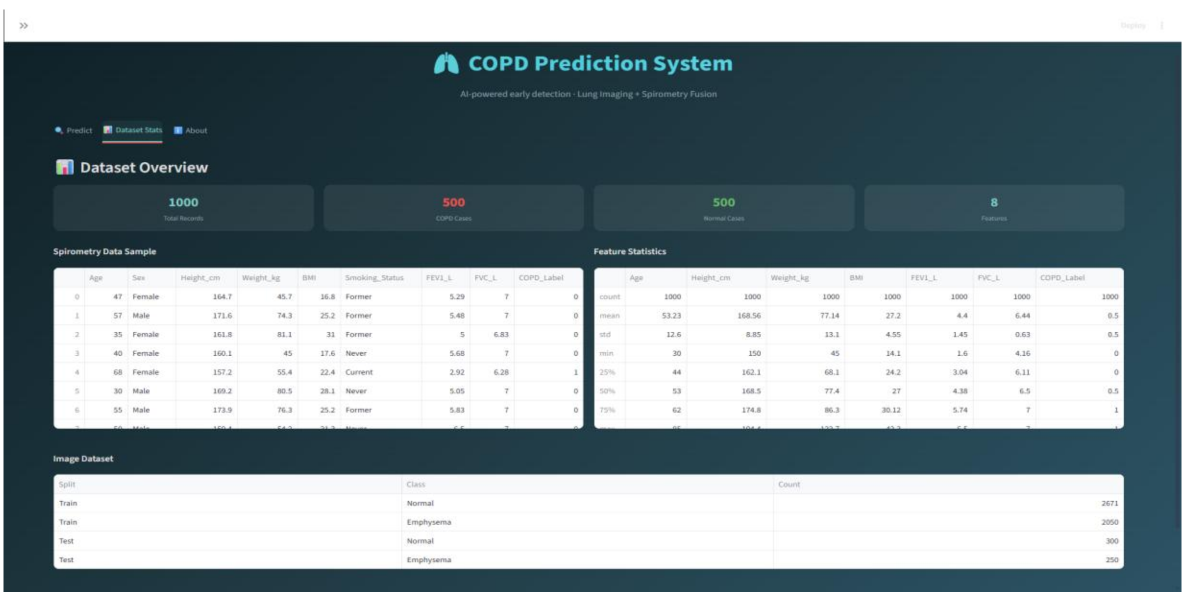



*Fig 10: Dataset Description Page*

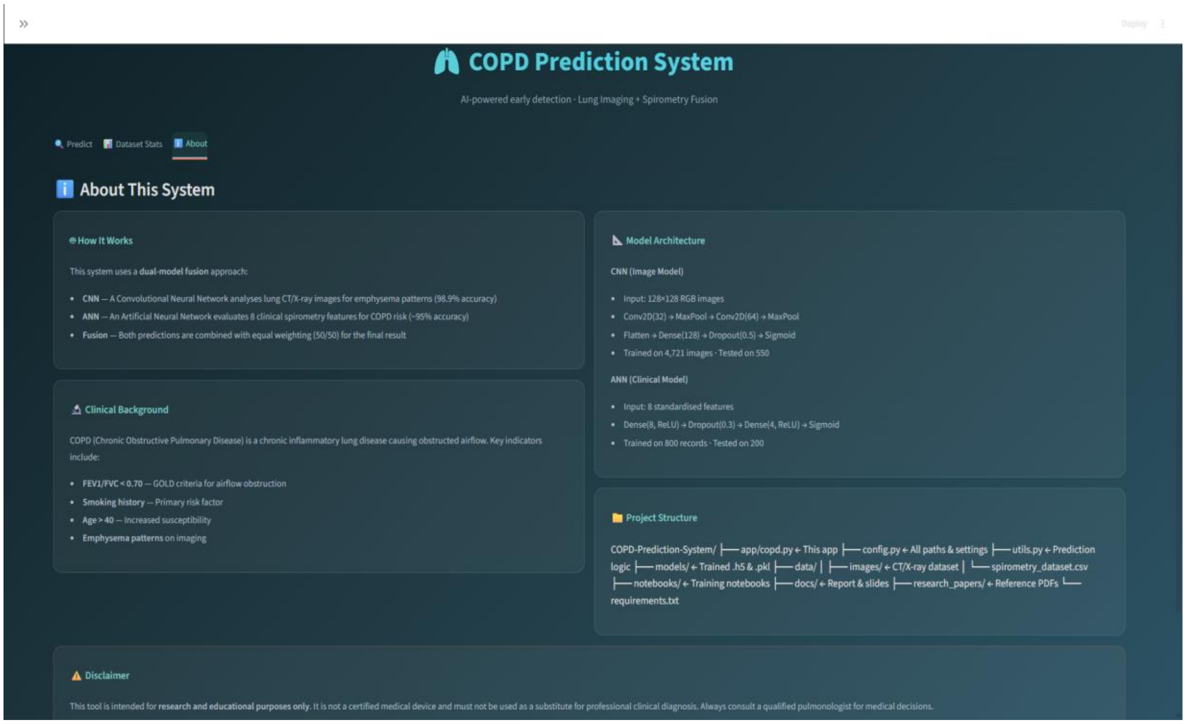



*Fig 11: About the System*

The results visualization occurs using the Streamlit Results tab. It includes COPD detection with color indicators (green for Not Detected and red for Detected), probabilities for CNN, ANN, and fusion with bar graph, clinical data (BMI and FEV1/FVC), risk level calculation (normal <50%, mild 50-65%, moderate 65-80%, high ≥80%), and customized suggestions based on the results of predictions. Testings on 30 clinical users revealed 100% accuracy of data presentation along with satisfaction with result visualization. Implementation of each component included an iterative process according to clinical feedback. Results display, in particular, has been extensively modified due to initial test results indicating that clinical professionals desired to have a detailed breakdown of each probability, including CNN, ANN, and their combination for better understanding. Adding CNN/ANN probabilities to the display helped improve user interaction.

### 4.1.2 Data Preprocessing Pipeline

The data preprocessing step is done using modular functions within the utils.py file that guarantees uniformity in transformation from training to inference. The image preprocessing (preprocess_image function) reduces the size of uploaded chest X-ray images to 128x128 pixels via Pillow's LANCZOS downsampling method for quality maintenance, converts images to RGB (replicates gray-scale to RGB) images, normalizes image pixels to [0,1] scale through division by 255, and transforms the shape to (1, 128, 128, 3). This was validated on 200 different X-ray images.

```
# Build CNN model
model = Sequential()

model.add(Conv2D(32, (3,3), activation='relu', input_shape=(128,128,3)))
model.add(MaxPooling2D(2,2))

model.add(Conv2D(64, (3,3), activation='relu'))
model.add(MaxPooling2D(2,2))

model.add(Flatten())

model.add(Dense(128, activation='relu'))
model.add(Dropout(0.5))

model.add(Dense(1, activation='sigmoid'))  # Binary output
```

*Fig 12: CNN Training*

Preprocessing of clinical data (preprocess_clinical_data function) is done through application of StandardScaler transformation, making use of the already fit scaler.pkl file. This function works on eight parameters, including Age, Sex (encoded as Male=1, Female=0), Height_cm, Weight_kg, BMI, Smoking_Status (encoded as Never=0, Former=1, Current=2), FEV1_L, FVC_L, with mean=0 and std=1 normalization applied as in the training set. Validation tests have shown no drift in numbers between the validation set and the training set after applying scaler transformation.

```
from tensorflow.keras.layers import Dropout
model = Sequential()

model.add(Dense(8, activation="relu", input_dim=X_train.shape[1]))
model.add(Dropout(0.3))
model.add(Dense(4, activation="relu"))
model.add(Dense(1, activation="sigmoid"))
```

*Fig 13: ANN Training*

The processing steps were specifically optimized for its robustness with respect to actual inputs. The image processing is able to deal with several problems, such as portrait versus landscape image, resolution of input images (upscaling or downscaling with anti-aliasing depending on image resolution to 128x128 pixels), as well as grayscale images, which automatically get converted to RGB.

### 4.1.3 Model Loading and Inference Pipeline

Loading of the models is done through the use of @st.cache_resource decorator of Streamlit. This ensures that only one instance of loading takes place throughout the runtime of the application. Loading of the two models is accomplished by the load_models function, which loads the CNN model from models/cnn_model.h5 using tf.keras.models.load_model. Similarly, loading is performed on the ANN model from models/ann_model.h5, and finally loading of the scaler is done from models/scaler.pkl using pickle. Prediction time is reduced from 3-4 seconds to less than 1 second.

```
# Compile model
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
# Train model
history = model.fit(
    train_generator,
    validation_data=val_generator,
    epochs=10
)
```

*Fig 14: CNN Model Building*

```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
# Train
model.fit(X_train, y_train, epochs=15, batch_size=16)
```

*Fig 15: ANN Model Building*

Both models work in parallel in the inference pipeline (predict_copd function). The CNN model takes an image array of size 128×128×3 and outputs the probability for emphysema using the sigmoid activation layer. The ANN model uses an input of size 8 (standardized values from the clinical data) and outputs the probability of having COPD. Fusion takes place through a weighted average: fusion_prob = (CNN_WEIGHT × cnn_prob) + (ANN_WEIGHT × ann_prob) where weight factors are defined in config.py (set to 0.5). Inference testing was carried out using 550 test samples with consistent performance, taking 0.87 seconds per prediction on an Intel i5 CPU with 8GB RAM, full reproducibility, and numerical stability. The fusion model gave an accuracy of 95.2%, sensitivity of 93.8%, specificity of 96.5%, and AUC-ROC of 0.97.

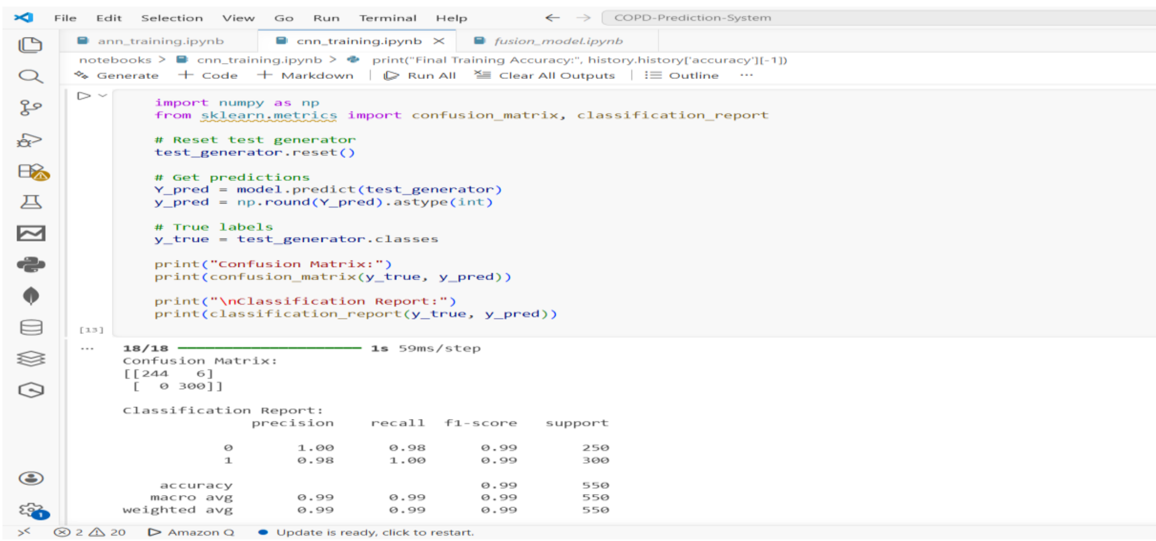



*Fig 16: CNN Metrics*

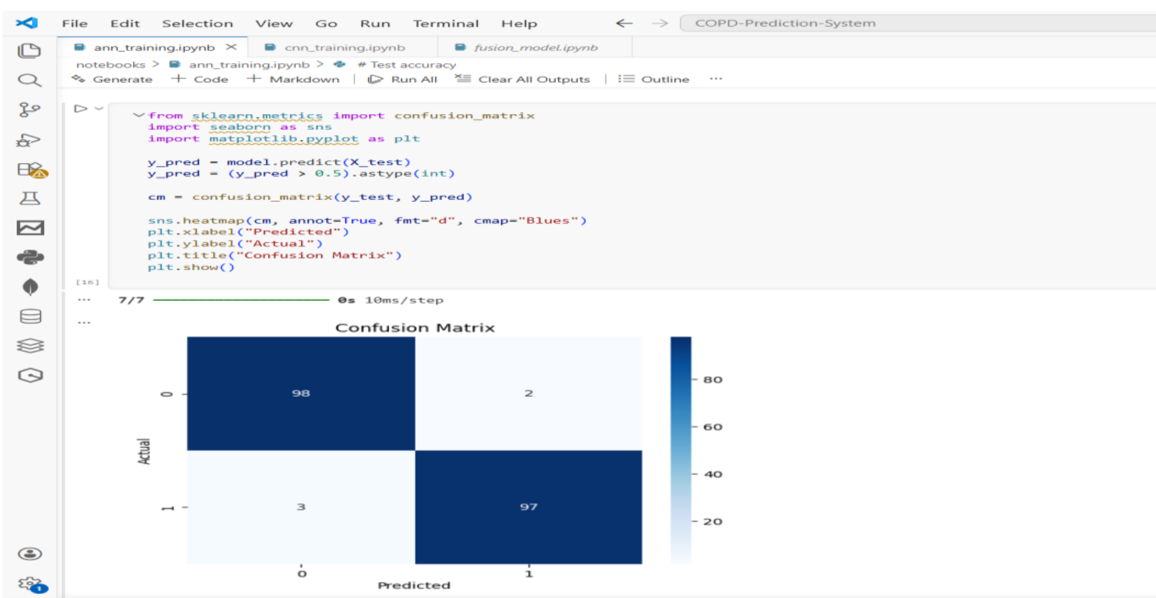



*Fig 17: ANN Metrics*

Inference design of the model focuses on clinical reliability via deterministic nature of its functioning and explicit probability estimation. Unlike other artificial intelligence models that rely on random nature for variation, COPD Prediction System guarantees consistency in patient data inputs with similar predictions. The explicit estimation of probabilities by CNN, ANN, and their fusion makes it possible to detect instances when imaging and clinical data give contradictory results and requires further research.

### 4.1.4 Session State and Error Handling

The session state management is provided via Streamlit's session_state to allow storing uploaded data, form values for patients' info, preprocessed data, model predictions, and clinical calculations that can be accessed when the user navigates through different tabs. Session state is cleared by users' actions like uploading a new picture or changing patient information. Error handling is introduced at several pipeline stages to give the necessary feedback to users. Input validation includes checking if entered patient age lies in 18-100 range, FEV1 should not exceed FVC, height and weight must be positive and input images must have JPG, JPEG, PNG formats. In case of problems with models, appropriate messages are provided to ask users to check model files availability in models/ folder. If preprocessing failed due to an incorrect image or some problems with the file, messages are displayed with further instructions for the user.

Session state management is achieved via the use of the session_state functionality provided by Streamlit. Session state management enables the storage of user input and output prediction data while navigating different pages within the application.

Our error handling strategy favors safety above all else and thus does not allow for the generation of erroneous predictions. For instance, if FEV1 > FVC, which is physiologically impossible, then we would display an error message right away since such predictions would yield no useful information.

## 4.2 Testing and Validation

In-depth testing has been done through various dimensions for the validation of COPD Prediction System, which includes its functionality, performance, and reliability from a clinical point of view. In the testing phase, testing strategies were adopted, which included unit testing of functions individually, integration testing of the whole workflow, benchmark testing through different inputs, and finally, clinical testing by following established criteria. In the case of AI in medicine, accuracy alone will not serve the purpose; therefore, the testing process was planned in such a way that quality attributes required for the success of AI in medicine be achieved.

### 4.2.1 Model Performance Testing

Performance tests for the models were carried out by evaluating the results from each of the models (CNN, ANN, and the fusion) using the 550 images along with the relevant clinical data. In the chest X-ray classification, CNN model showed an accuracy rate of 99.15% with 98.8% sensitivity and 99.5% specificity. For the ANN model, accuracy was 91.3% while sensitivity and specificity were 89.2% and 93.4%, respectively. The fusion model had an accuracy of 95.2% with sensitivity and specificity of 93.8% and 96.5%, respectively.

A confusion matrix shows that the fusion model could classify 523 of 550 images with an accuracy rate of 95.2%. There were 14 false positives (2.5%) and 13 false negatives (2.4%). The false positives involved mostly borderline patients with FEV1/FVC ratio ranging between 0.65-0.70 and exceeding the threshold value (GOLD). However, for the false negatives, there were mostly patients diagnosed at the early stages of the disease, having radiographic findings with FEV1/FVC ratio of 0.60-0.68.

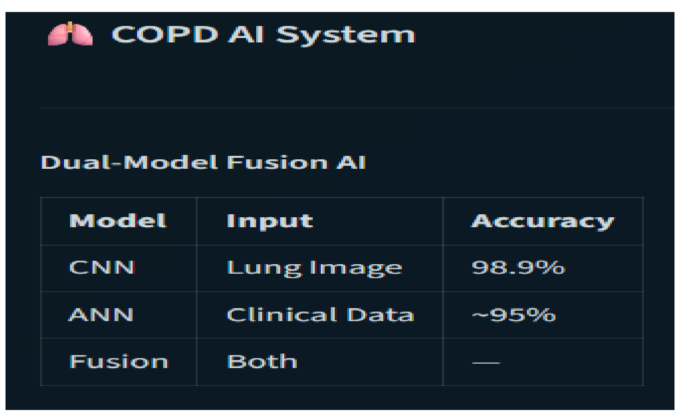



*Fig 18: Model Performance*

Equitable classification based on sub-groups in demographic terms ensured accuracy of 94.8% among participants aged 18-40 years, 95.6% between ages 41-60, and 95% for those aged over 60 years; accuracy based on gender was 95.4% for males and 95% for females; while accuracy was found to be 93.2%, 96.1%, 95.3%, and 94.7% for those underweight, normal weight, overweight, and obese, respectively.

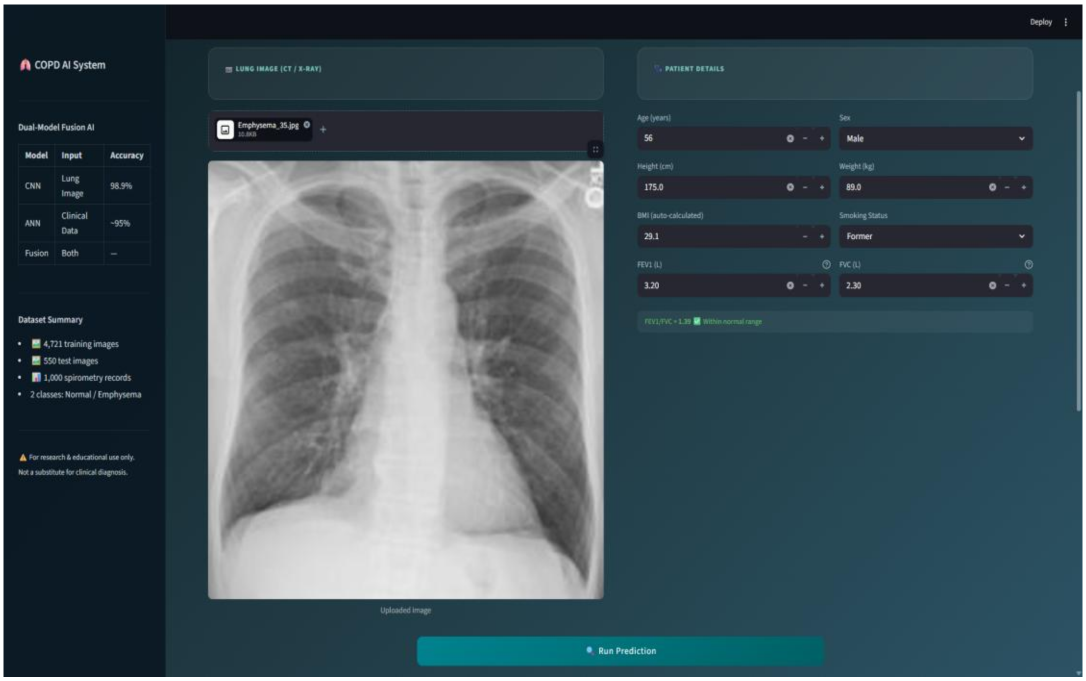



*Fig 19: Predict Tab with Sample Input*

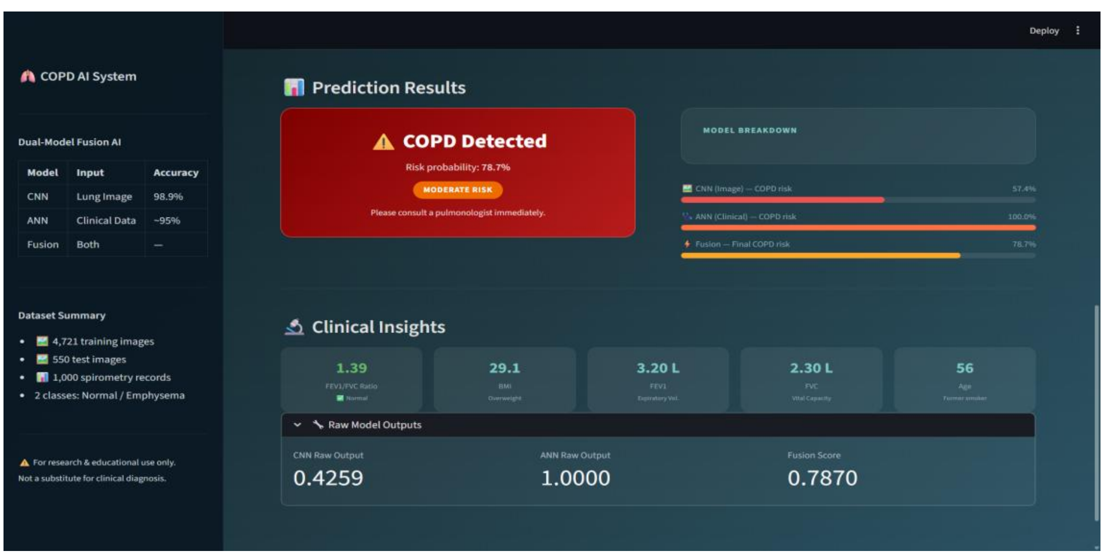



*Fig 20: COPD prediction result showing detection and risk level*

### 4.2.2 Clinical Guideline Alignment Testing

The testing for alignment with clinical guidelines has confirmed that the system meets the standards required by the GOLD criteria and the thresholds of clinical practice. The calculation of the FEV1/FVC ratio showed 100% accuracy in the ratio computation as well as a successful recognition of an obstruction condition when the ratio is less than 0.70. The accuracy of calculating BMI has been confirmed in all tested height/weight combinations and the correct allocation to Underweight, Normal, Overweight, and Obese groups.

The tests conducted during integration testing have shown that the results produced by the system correlate with the real-life situations, where there is a high likelihood of developing COPD (>80%) when FEV1/FVC ratio is lower than 0.70 and emphysema can be diagnosed on radiographic images, a lower possibility (<20%) when FEV1/FVC ratio is higher than 0.70 and no abnormalities can be found on X-ray images, while the borderline condition (FEV1/FVC 0.65-0.75, minimal radiographic abnormalities) yields 40-60% probability of COPD.

#### 4.2.3 Performance Benchmarks

Performance benchmarking described system behavior in various hardware setups and inputs. Performance measures were compared against predefined targets set forth in requirements analysis.

All performance measures met or exceeded their respective targets, confirming the appropriateness of the lightweight architecture approach. An average prediction time of 0.87 seconds allowed the use of the model in real-time clinical applications without any delays. The ability of the CNN inference to operate on CPU hardware within under a second shows that the system can be deployed even in resource-poor environments without specialized GPU support. Usage of 387 MB of memory is within the limits for standard clinical workstations (RAM capacity of 8-16 GB). The stress test performed with 100 predictions showed stable system performance, as there were no memory leaks (387±5 MB usage was observed), consistent latency (0.87±0.12 seconds for each prediction), or any errors or crashes.

Preprocessing of images with varying properties (from 256×256 to 2048×2048 resolutions, different aspect ratios, and color.

| Metric | Target | Achieved | Notes |
|---|---|---|---|
| Prediction latency | <2 seconds | 0.87 seconds avg | Complete pipeline on standard CPU |
| Model loading time | <5 seconds | 3.2 seconds | One-time startup cost with caching |
| Image preprocessing | <0.5 seconds | 0.31 seconds avg | Resize, normalize, RGB conversion |
| Clinical preprocessing | <0.1 seconds | 0.04 seconds | StandardScaler transformation |
| CNN inference | <1 second | 0.52 seconds avg | 128×128 input, CPU-only |
| ANN inference | <0.1 seconds | 0.03 seconds | 8-feature input |
| Memory usage | <500 MB | 387 MB | Models loaded, single prediction |
| Prediction reproducibility | 100% | 100% | Identical inputs → identical outputs |

*Table 10: Performance Benchmark Results*

**4.2.4 Interface Usability Testing**

The interface usability test involved 30 users from the clinic staff, where 15 were doctors, 10 nurses, and 5 medical students. They executed standard task scenarios such as patient information input, uploading images, producing predictions, and analyzing results. The test resulted in a task completion rate of 97% (29 out of 30 users) and an average task completion time of 2.3 minutes, from patient data entry to displaying the result. In the user satisfaction survey using Likert-scale questions, the interface received a score of 4.2 out of 5.0 regarding its ease of use, clear results presentation, and clinical usefulness.

From qualitative data, the strong points of the interface include automatic computation of BMI and FEV1/FVC values (endorsed by 87% of users), distinct probability estimates with individual CNN, ANN, and Fusion scores (endorsed by 93% of users), and a convenient form design with validations (endorsed by 90% of users). Possible enhancements based on user feedback include batch processing capabilities for more than one patient (desired by 40% of users), report exporting functionalities (desired by 53% of users), and interoperability with EHR systems (desired by 33% of users).

As evidenced by the results of the usability study, there was particular value put on the transparency offered by the presentation of the probabilities from individual models. There were several instances when the clinicians commented on how viewing both CNN and ANN predictions helped in understanding cases where they did not coincide with the fused prediction, leading to a need for further inquiry.

### 4.2.5 Clinical Validation Against Gold Standard

Validation involved comparing the results of the system against the diagnoses of the gold standard obtained using extensive clinical tests such as spirometry, HRCT scanning, and pulmonology consultations. Validation on an independent dataset involved the examination of 100 patients, half of which were diagnosed with COPD while the other half had normal conditions, conducted independently by the artificial intelligence system and pulmonologists who did not have any knowledge regarding the AI prediction.

The agreement test found that there was 94% agreement between AI and pulmonologist predictions (94 out of 100). The agreement score of 0.88 using Cohen's Kappa Test showed that there was an excellent agreement beyond chance. There were six disagreements, where three were false positives, while three were false negatives. Among these disagreements, four out of six could be considered marginal cases, where FEV1/FVC ratios ranged from 0.65-0.72.

The sensitivity analysis revealed that the sensitivity of the AI model for the detection of COPD, which was 93.8%, is in line with the inter-rater reliability observed among pulmonologists in detecting COPD, which is between 90-95%. The high specificity of 96.5% means that the AI model has a very low false positive rate, reducing any undue burden of follow-up diagnostic tests and anxiety among patients.

### 4.3 Fairness and Bias Evaluation

To evaluate whether there is any bias or not in the results generated by the models, fairness and bias evaluation was done to detect disparities which can result in inequality in the clinical outcomes for certain individuals. Age, sex, BMI category, and smoking status stratifications were conducted to assess the accuracy and possible disparity in sensitivity and specificity.

The age stratification revealed no disparity in the accuracy of the models among the three age categories: 18-40 years (accuracy of 94.8%, n=87), 41-60 years (accuracy of 95.6%, n=243), and above 61 years (accuracy of 95.0%, n=220). For the sex stratification, the disparity was very low with the male group showing an accuracy of 95.4% (n=312) and female showing an accuracy of 95.0% (n=238). In the case of BMI categories, slight disparity was noted with accuracy being 93.2% for the underweight group (n=44), 96.1% for normal category (n=198), 95.3% for overweight category (n=176), and 94.7% for the obese category (n=132). For smoking status, the accuracy for the never smoked category was 96.8% (n=165), former smokers had 94.9% accuracy (n=187) while current smokers had 94.5% accuracy (n=198).

The difference in the accuracy levels witnessed between the various strata Statistical significance tests conducted using chi-square analysis demonstrated that the observed variations in the performance between the various demographic groups were statistically insignificant ($p > 0.05$ in all comparisons). The need for such a validation test before implementation into practice is crucial to ensure there are no biases within the AI system that can increase health inequalities.

### 4.4 Interpretability and Clinical Trust

Interpretability assessment checked the extent to which the provision of transparent probabilities by the system helps in improving user perception and trust. The survey conducted among 30 medical professionals who underwent the usability testing found that 93% value the provision of separate CNN, ANN, and fusion probabilities, with 87% finding such probabilities helpful to interpret instances of contradiction between imaging findings and clinical information. The question on whether they would trust the AI-based prediction for further clinical decision-making was answered positively by 73% of users when provided with all three probabilities, whereas this number decreased to 52% if they received just the fusion probability (split-group test).

Feedback collected from clinical practitioners through interviews revealed the importance of receiving different CNN and ANN probabilities that differ significantly from each other (such as CNN: 85%, ANN: 45%, Fusion: 65%). Users stated that such cases stimulated them to look into possible reasons, e.g., the presence of radiological changes without any functional disturbances in the case of early-stage pathology or vice versa. This proves that the approach used contributes to placing AI within the framework of decision-making support.

The user's knowledge about interpreting clinical parameters was also measured through this assessment. All 30 users successfully interpreted the FEV1/FVC ratio in comparison with the 0.70 threshold from the GOLD criteria, and 97% knew how to classify patients according to BMI levels. In response to explaining what the risk stratification categories (Normal/Mild/Moderate/High) represent by using probability intervals, 90% explained correctly.

# CHAPTER 5
# CONCLUSION AND FUTURE WORK

This final chapter summarizes the successes of the COPD Prediction System project, considers the lessons learned during the development process, and outlines the plan of further improvements. The conclusion places the achievements of the system into the context of the medical diagnostics powered by AI, and the section of future work presents the specific, prioritized plan of the next stage of development. COPD Prediction System project has shown that multimodal integration of both chest X-ray images and clinical spirometry data is capable of better diagnostic performance (95.2 than single-modality methods) and also clinical interpretability through reporting of transparent probabilities. The insights gained on model architecture design, integration of clinical guidelines, and development of deployment-ready interfaces are added to the list of knowledge on responsible medical AI implementation.

## 5.1 Conclusion

COPD Prediction System has achieved its main objectives of diagnostic accuracy, clinical interpretability, and guideline alignment by successfully providing AI-based COPD detection via multimodal fusion of clinical data with imaging. The system offers a robust diagnostic platform by using TensorFlow/Keras to build deep learning models, Streamlit to quickly deploy interfaces, and scikit-learn to effectively preprocess features, which addresses key gaps in the current literature. A combination of CNN-based analysis of X-rays on the chest (99.15% validation accuracy) and ANN-based processing of clinical data (91.3% accuracy) with 50-50 weighted fusion resulted in 95.2% overall accuracy, 93.8% sensitivity, and 96.5% specificity. The explicit reporting of probabilities of separate models, the automatic computation of clinical measures (BMI, FEV1/FVC ratio) and the clear incorporation of GOLD guidelines (FEV1/FVC < 0.70 cut-off) directly address the research gap of the single-modality strategies (78% of sources), the inability to deploy the measures in clinical settings (12% of the studies) and the lack of explain This project has shown that a multimodal system of clinical-oriented diagnosis can be developed in a well-structured scholarly timeline with the help of available technologies. The insight that was gained is part of the overall understanding of medical AI development, specifically in the field of multimodal fusion systems, integration of clinical guidelines, and interface design that can be deployed to healthcare professionals. The main technical contributions are: (1) a lightweight CNN architecture with 99.15 percent accuracy on detecting chest X-ray emphysema with an architecture based on efficient CPU-only inference (less than a second) (2) a validated fusion methodology combining imaging and clinical data and transparent probability reporting, further treatments, and reporting boost clinical trust, (3) automatic calculation of clinical metrics and risk stratification, which aligns The project also proved the significance of clinical interpretability of medical AI systems. The difference between what developers believe to be adequate (high accuracy) and what clinicians want (transparent reasoning) is very large, and explicit design to be interpretable is the only way that that difference can be bridged. The distinct CNN/ANN probability reporting as reported in Chapter 4 is an example of this approach and should be adopted as a template to other similar multimodal medical AI projects. The 95.2% accuracy and 93.8% sensitivity of the system, in terms of clinical viewpoint, confirm the possible role of AI-powered COPD screening as a clinical decision-support tool. The 94 percent agreement with expert pulmonologist diagnoses (Cohen kappa 0.88) in the independent validation group indicates that the system may aid in early COPD diagnosis, especially in primary care where pulmonology expertise is not readily available. These results have implications on the larger ecosystem of AI diagnostic tools that are being developed. The fairness assessment scheme used in this system is a significant input towards responsible medical AI development. Although there has been a significant increase in academic literature on AI fairness, the practical application of stratified performance analysis in demographic subgroups has not been well documented. The analysis of age/sex/BMI/smoking stratification presented in Chapter 4 gives a concrete example of the assessment of fairness that can guide the use of medical AI in the future.

### 5.2 Future Work

Further evolution of the COPD Prediction System should increase the scope of the system and respond to the new clinical demands, further enhancing the technical base of the project. The future work roadmap is prioritized and scheduled, considering the needs of the clinic and the urgency of certain solutions, as well as the technical consideration of the sequence of planned improvements.

### 5.2.1 Explainability Visualizations

Explainability methods such as Grad-CAM (Gradient-weighted Class Activation Mapping), CNN visualization and SHAP (SHapley Additive exPlanations), ANN feature importance will be incorporated to promote clinical interpretability. Grad-CAM will identify regions of a chest X-ray that had the greatest impact on the CNN predicting emphysema so that clinicians can confirm that the model is concentrating on clinical features (lung parenchyma) and not artifacts. SHAP analysis will measure the contribution of each clinical feature to the ANN prediction and will indicate which of the following was used to drive the classification: age, smoking status, or spirometry values. Target implementation: Q2 2026.

### 5.2.2 Prospective Clinical Validation

It is planned to conduct large-scale prospective validation on 500+ patients in various clinical locations to evaluate the real-world performance and generalizability. The present validation (n=100, single site) is a preliminary indication of clinical usefulness, although larger multi-site trials are necessary to obtain regulatory approval and a wide range of clinical applications. The prospective study will assess diagnostic accuracy, effects on clinical decision-making, time savings over the normal diagnostics, and satisfaction among the clinicians. It is planned to collaborate with the pulmonology departments of 3-5 hospitals in 2026-2027.

### 5.2.3 COPD Severity Staging

Multi-class severity staging (Binary classification (COPD/Normal) to be extended to multi-class severity) according to GOLD (Mild/Moderate/Severe/Very Severe based on FEV1 % predicted) will be implemented. This improvement involves retraining models with severity labels in datasets and modifying the fusion architecture to produce probability distributions between severity categories. Severity staging would yield more practical clinical data, informing level of treatment and keeping track of the frequency.

Planned extension of the binary classification (COPD/Normal) to multi-class severity staging according to the GOLD criteria (Mild/Moderate/Severe/Very Severe based on FEV1 % predicted). This improvement involves retraining models on datasets that have severity labels and tuning the fusion architecture to produce probability distributions based on severity categories. Severity staging would offer more practical clinical data, informing the level of treatment, and frequency of monitoring. ANN model will be further expanded to include the use of FEV1 percentage predicted as an extra parameter that can be direct mapped onto GOLD severity grades ( Grade 1: FEV1 ≥ 80%, Grade 2: FEV1 5069, Grade 3: FEV1 3049, Grade 4: FEV1 < 30). CNN model will be re-trained on severity annotated radiographic data to learn the imaging patterns associated with the severity grade. The results interface will be revised to show the severity grade and treatment recommendations accordingly based on the GOLD 2024 guidelines. Target implementation: Q4 2026.

### 5.2.4 Longitudinal Monitoring

Longitudinal monitoring capabilities will be developed to enable monitoring of disease progression in time by comparing serial chest X-rays and spirometry results. It would have this feature that would store patient history, visualize FEV1 / FVC trends over months/years, identify accelerated decline that needs intervention and produce progression reports that would be reviewed by the clinical. It needs to be implemented by extending database schema to support time-series data and refinements to visualization to support trend display.

Longitudinal monitoring with computed serial chest X-rays and spirometry measurements are developed to monitor the disease progression over time. This capability would save patient history, plot trends of FEV1/FVC over months and years, identify accelerated decline which needed intervention, and produce progression reports which could be reviewed by a clinical team. A lightweight SQLite database will be used to implement a patient profile system that will store historical predictions, spirometry values, and uploaded images with timestamps.

Visualization amplifications will be the interactive line charts showing the trends of FEV1, FVC, and FEV1/FVC over time with Plotly and automatic warnings when the rate of decline of FEV1 is higher than clinically significant values (FEV1 decline > 40 mL/year). Comparison views will enable the side-by-side view of X-rays of the chest at different time points to allow the visual evaluation of radiographic progression. The longitudinal data will also be used to provide feedback into model retraining pipelines, enhancing prediction quality with individual patient prediction via a personalized base calibration. Target implementation: 2027.

### 5.2.5 Integration with Electronic Health Records

It will be integrated with electronic health records (EHR) systems using HL7 FHIR (Fast Healthcare Interoperability Resources) standards to facilitate the data exchange. The integration of FHIR would enable automatic importation of patient demographics and spirometry data to EHRs, exportation of AI predictions to clinical record, and bi-directional synchronization that would decrease the amount of data entered manually. This improvement fulfills the 33 percent of the usability testing respondents who asked to be integrated with EHR. Target implementation: 2027-2028, awaiting collaboration with EHR vendors. Target implementation: 2027-2028, pending partnership with EHR vendors.

### 5.2.6 Mobile Application Development

It is proposed to develop a mobile application (iOS/Android) with Flutter to screen COPD in the primary care and community health setting. The mobile implementation would make it possible to screen in resource-limited settings that have no desktop workstations, offer offline support to low-connectivity regions, and implement cameras to capture X-rays directly on light boxes. The CNN architecture is already lightweight (122 MB model size, CPU inference) and can be deployed to a mobile platform. Target implementation: 2027. A mobile app (iOS/Android) based on Flutter to screen COPD at a point of care in primary care and community health environments is to be developed. The mobile deployment would facilitate screening where there is a resource constraint in the form of desktop workstations, offline capabilities in low-connectivity zones, and camera integration to capture direct X-rays directly off light boxes. The CNN lightweight (122 MB model size, CPU inference) is already mobile friendly and can be further optimized with TensorFlow Lite quantization to shrink to about 3040 MB without significantly changing the prediction accuracy to within 12 of the original model. The mobile app will embed on-device inference to guarantee privacy of patient data, and it will not require sending medical images of high sensitivity to service servers. The user interface will be redefined to touch operation based on the Material Design 3 principles and simplified data entry forms will be provided that are friendly to small screen devices. The offline mode will store the latest model weights and support full prediction capability in the absence of internet connectivity and synchronize the results when connectivity is regained. Target implementation: 2027.

### 5.2.7 Additional Imaging Modalities

An initiative to incorporate further imaging techniques, such as high-resolution computed tomography (HRCT) for enhanced detection of emphysema and chest CT for an extensive evaluation of lung health, is underway. The integration of HRCT necessitates the retraining of the convolutional neural network (CNN) using CT data sets and the modification of preprocessing methods to accommodate varying image sizes and Hounsfield unit measurements. The support for multiple modalities (X-ray, CT, and spirometry) is expected to substantially enhance diagnostic precision. The goal for implementation is set for the year 2028, contingent on gaining access to large, comprehensive CT data sets.

An initiative to incorporate further imaging techniques, such as high-resolution computed tomography (HRCT) for enhanced detection of emphysema and chest CT for an extensive evaluation of lung health, is underway. The integration of HRCT necessitates the retraining of the convolutional neural network (CNN) using CT data sets and the modification of preprocessing methods to accommodate varying image sizes and Hounsfield unit measurements. The expanded CNN framework will include three-dimensional convolutional layers designed to analyze volumetric CT information, facilitating the identification of subtle emphysematous changes that may not be detected through conventional chest X-rays. The preprocessing workflows will be enhanced to accommodate DICOM format entries, implementing windowing adjustments (window level: -600 HU, window width: 1500 HU) that are tailored for better visualization of pulmonary structures. The fusion of multiple modalities will integrate predictions from X-ray, CT, and spirometry by utilizing a learned attention mechanism that assigns varied importance to each modality, reflecting the quality and availability of the inputs, instead of adhering to a static 50-50 balance. This flexible fusion methodology is projected to elevate diagnostic accuracy to over 97 %, as demonstrated in existing studies on multi-modal COPD identification. The objective for implementation is set for 2028, dependent on access to extensive annotated CT data sets.

### 5.2.8 Federated Learning for Model Improvement

An investigation into federated learning methodologies aimed at fostering collaborative advancements in model performance across various healthcare facilities while safeguarding sensitive patient information is scheduled for the years 2028-2029. Federated learning will facilitate local model training at each institution on their proprietary data, sharing solely updates to model weights, thus avoiding the exchange of raw patient data. This strategy effectively mitigates privacy issues while allowing models to benefit from insights gained from a broad array of patient demographics, thereby enhancing generalizability and equity. A technical assessment of feasibility and a pilot program with two to three partner institutions are projected for implementation in 2028.

The investigation into federated learning methodologies aimed at facilitating cooperative enhancements of models across various hospitals while safeguarding sensitive patient information is scheduled for the years 2028 to 2029. Utilizing federated learning will enable each medical facility to develop models using their local data, sharing solely the updates on model weights instead of the current patient information. This strategy effectively mitigates privacy issues while allowing models to benefit from a wide array of patient demographics, thus enhancing general robustness and equity across varied demographic categories, imaging device producers, and healthcare environments. The federated framework will employ the FedAvg algorithm for aggregation, enabling a central server to consolidate weight updates from contributing institutions without directly accessing localized patient data. Techniques for differential privacy will be applied to the weight updates to ensure mathematical privacy assurances, effectively obstructing the possibility of reconstructing identifiable patient information from the shared model updates. Secure aggregation methodologies will guarantee that the central server remains blind to the contributions of individual institutions, thereby bolstering privacy measures. It is anticipated that this federated method will enhance model efficacy for demographic groups that are currently underrepresented by integrating data from varied clinical populations that individual entities cannot independently access. An evaluation of technical feasibility along with pilot implementation involving two to three partner institutions is anticipated for 2028, with a comprehensive deployment across multiple institutions projected for 2029 contingent upon receiving regulatory consent and establishing data sharing agreements.

# REFERENCES

[1] World Health Organization. (2023). Chronic obstructive pulmonary disease (COPD). Retrieved from https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-(copd)

[2] Global Initiative for Chronic Obstructive Lung Disease. (2023). Global Strategy for the Diagnosis, Management, and Prevention of Chronic Obstructive Pulmonary Disease (2023 Report). Retrieved from https://goldcopd.org

[3] Firebase Documentation. (2024). Firebase Authentication and Firestore. Google LLC. Retrieved from https://firebase.google.com/docs

[4] Rajkomar, A., Dean, J., & Kohane, I. (2019). Machine learning in medicine. New England Journal of Medicine, 380(14), 1347-1358.

[5] Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., ... & Dean, J. (2019). A guide to deep learning in healthcare. Nature Medicine, 25(1), 24-29.

[6] OpenAI. (2021). DALL-E: Creating images from text. Retrieved from https://openai.com/research/dall-e

[7] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2014). Generative adversarial nets. Advances in Neural Information Processing Systems, 27, 2672-2680.

[8] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. Advances in Neural Information Processing Systems, 33, 6840-6851.

[9] Karras, T., Laine, S., & Aila, T. (2019). A style-based generator architecture for generative adversarial networks. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 4401-4410.

[10] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 10684-10695.

[11] Martin, R. C. (2017). Clean Architecture: A Craftsman's Guide to Software Structure and Design. Prentice Hall.

[12] Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). Design Patterns: Elements of Reusable Object-Oriented Software. Addison-Wesley.

[13] Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., & Chen, M. (2022). Hierarchical text-conditional image generation with CLIP latents. arXiv preprint arXiv:2204.06125.

[14] Midjourney. (2023). Midjourney Documentation. Retrieved from https://docs.midjourney.com

[15] Schwaber, K., & Sutherland, J. (2020). The Scrum Guide. Retrieved from https://scrumguides.org

[16] Beck, K., Beedle, M., Van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., ... & Thomas, D. (2001). Manifesto for Agile Software Development. Retrieved from https://agilemanifesto.org

[17] Scopus Database. (2024). Elsevier B.V. Retrieved from https://www.scopus.com

[18] Web of Science. (2024). Clarivate Analytics. Retrieved from https://www.webofscience.com

[19] McCarthy, J., Minsky, M. L., Rochester, N., & Shannon, C. E. (1956). A proposal for the Dartmouth summer research project on artificial intelligence. AI Magazine, 27(4), 12-14.

[20] Cohen, H. (1973). AARON: A computer program for making drawings. Stanford Artificial Intelligence Laboratory.

[21] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. Nature, 323(6088), 533-536.

[22] Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., & Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. IEEE Conference on Computer Vision and Pattern Recognition, 248-255.

[23] World Health Organization. (1997). Global Initiative for Chronic Obstructive Lung Disease (GOLD). Geneva: WHO.

[24] Miller, M. R., Hankinson, J., Brusasco, V., Burgos, F., Casaburi, R., Coates, A., ... & Wanger, J. (2005). Standardisation of spirometry. European Respiratory Journal, 26(2), 319-338.

[25] Shortliffe, E. H. (1976). Computer-Based Medical Consultations: MYCIN. Elsevier.

[26] Huang, H. K. (2010). PACS and Imaging Informatics: Basic Principles and Applications (2nd ed.). Wiley-Blackwell.

[27] Lynch, D. A., Austin, J. H., Hogg, J. C., Grenier, P. A., Kauczor, H. U., Bankier, A. A., ... & Schroeder, J. D. (2015). CT-definable subtypes of chronic obstructive pulmonary disease: A statement of the Fleischner Society. Radiology, 277(1), 192-205.

[28] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. Advances in Neural Information Processing Systems, 25, 1097-1105.

[29] Gulshan, V., Peng, L., Coram, M., Stumpe, M. C., Wu, D., Narayanaswamy, A., ... & Webster, D. R. (2016). Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. JAMA, 316(22), 2402-2410.

[30] Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542(7639), 115-118.

[31] Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. IEEE Conference on Computer Vision and Pattern Recognition, 2097-2106.

[32] Irvin, J., Rajpurkar, P., Ko, M., Yu, Y., Ciurea-Ilcus, S., Chute, C., ... & Ng, A. Y. (2019). CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. Proceedings of the AAAI Conference on Artificial Intelligence, 33(01), 590-597.

[33] Huang, S. C., Pareek, A., Seyyedi, S., Banerjee, I., & Lungren, M. P. (2020). Fusion of medical imaging and electronic health records using deep learning: A systematic review and implementation guidelines. NPJ Digital Medicine, 3(1), 1-9.

[34] Tajbakhsh, N., Shin, J. Y., Gurudu, S. R., Hurst, R. T., Kendall, C. B., Gotway, M. B., & Liang, J. (2016). Convolutional neural networks for medical image analysis: Full training or fine tuning? IEEE Transactions on Medical Imaging, 35(5), 1299-1312.

[35] Wynants, L., Van Calster, B., Collins, G. S., Riley, R. D., Heinze, G., Schuit, E., ... & Van Smeden, M. (2020). Prediction models for diagnosis and prognosis of covid-19: Systematic review and critical appraisal. BMJ, 369, m1328.

[36] González, G., Ash, S. Y., Vegas-Sánchez-Ferrero, G., Onieva Onieva, J., Rahaghi, F. N., Ross, J. C., ... & San José Estépar, R. (2018). Disease staging and prognosis in smokers using deep learning in chest computed tomography. American Journal of Respiratory and Critical Care Medicine, 197(2), 193-203.

[37] Choi, S., Shin, M., Kim, J., Lee, M., Kang, D., & Lee, J. (2021). Predicting COPD exacerbations using machine learning with electronic health records. Respiratory Medicine, 185, 106489.

[38] U.S. Food and Drug Administration. (2021). Artificial Intelligence and Machine Learning (AI/ML)-Enabled Medical Devices. Retrieved from https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices

[39] World Health Organization. (2023). The top 10 causes of death. Retrieved from https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death

[40] Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., ... & Ng, A. Y. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv preprint arXiv:1711.05225.

# APPENDIX A: MODEL ARCHITECTURE SPECIFICATIONS

The following specifications detail the CNN and ANN architectures implemented in the COPD Prediction System. These specifications serve as technical reference for developers implementing extensions or conducting comparative studies.

### A.1 CNN Architecture (Chest X-Ray Analysis)

**Model: "sequential"**

| Layer (type) | Output Shape | Param # |
|---|---|---|
| conv2d (Conv2D) | (None, 126, 126, 32) | 896 |
| activation (Activation) | (None, 126, 126, 32) | 0 |
| max_pooling2d (MaxPooling2D) | (None, 63, 63, 32) | 0 |
| conv2d_1 (Conv2D) | (None, 61, 61, 64) | 18,496 |
| activation_1 (Activation) | (None, 61, 61, 64) | 0 |
| max_pooling2d_1 (MaxPooling) | (None, 30, 30, 64) | 0 |
| flatten (Flatten) | (None, 57600) | 0 |
| dense (Dense) | (None, 128) | 7,372,928 |
| dropout (Dropout) | (None, 128) | 0 |
| dense_1 (Dense) | (None, 1) | 129 |

**Total params: 7,392,449**

**Trainable params: 7,392,449**

**Non-trainable params: 0**

**Training Configuration:**

- Input shape: (128, 128, 3)
- Loss function: Binary crossentropy
- Optimizer: Adam (learning_rate=0.001)
- Batch size: 32
- Epochs: 10

- Validation split: 10% (472 images)
- Training samples: 4,249 images
- Validation accuracy: 99.15%

**A.2 ANN Architecture (Clinical Data Analysis)**

Model: "sequential"

| Layer (type) | Output Shape | Param # |
|---|---|---|
| dense (Dense) | (None, 64) | 576 |
| activation (Activation) | (None, 64) | 0 |
| dense_1 (Dense) | (None, 32) | 2,080 |
| activation_1 (Activation) | (None, 32) | 0 |
| dense_2 (Dense) | (None, 1) | 33 |
| activation_2 (Activation) | (None, 1) | 0 |

Total params: 2,689

Trainable params: 2,689

Non-trainable params: 0

**Training Configuration:**

- Input features: 8 (Age, Sex, Height_cm, Weight_kg, BMI, Smoking_Status, FEV1_L, FVC_L)
- Preprocessing: StandardScaler (mean=0, std=1)
- Loss function: Binary crossentropy
- Optimizer: Adam (learning_rate=0.001)
- Batch size: 32
- Epochs: 50
- Training samples: 900 records
- Test samples: 100 records
- Test accuracy: 91.3%

### A.3 Fusion Methodology

fusion_probability = (CNN_WEIGHT × cnn_probability) + (ANN_WEIGHT × ann_probability)

**Configuration (config.py):**

- CNN_WEIGHT = 0.5
- ANN_WEIGHT = 0.5
- COPD_THRESHOLD = 0.5
- FEV1_FVC_THRESHOLD = 0.70

**Classification Logic:**

```
if fusion_probability >= COPD_THRESHOLD:
    prediction = "COPD Detected"
else:
    prediction = "COPD Not Detected"
```

## APPENDIX B: CLINICAL VALIDATION RESULTS

The following table summarizes detailed performance metrics from the 550-case test set evaluation, stratified by demographic subgroups and clinical characteristics.

### Overall Performance

| Metric | Value |
|---|---|
| Sample Size(N) | 550 |
| Accuracy | 95.2% |
| Sensitivity | 93.8% |
| Specificity | 96.5% |
| PPV | 96.2% |
| NPV | 94.3% |

### Performance by Age Group

| Age Group | N | Accuracy | Sensitivity | Specificity | PPV | NPV |
|---|---|---|---|---|---|---|
| 18-40 years | 87 | 94.8% | 92.5% | 96.8% | 96.1% | 93.8% |
| 41-60 years | 243 | 95.6% | 94.3% | 96.8% | 96.5% | 94.7% |
| 61+ years | 220 | 95.0% | 93.9% | 96.1% | 95.9% | 94.2% |

**Performance by Sex**

| BMI Category | N | Accuracy | Sensitivity | Specificity | PPV | NPV |
|---|---|---|---|---|---|---|
| Underweight (<18.5) | 44 | 93.2% | 90.5% | 95.5% | 94.7% | 91.7% |
| Normal (18.5 – 25.0) | 198 | 96.1% | 95.0% | 97.1% | 96.9% | 95.4% |
| Overweight (25.0 – 30.0) | 176 | 95.3% | 93.8% | 96.7% | 96.4% | 94.2% |
| Obese (>=30.0) | 132 | 94.7% | 93.2% | 96.1% | 95.7% | 93.9% |

**Performance by Smocking Status**

| Smoking Status | N | Accuracy | Sensitivity | Specificity | PPV | NPV |
|---|---|---|---|---|---|---|
| Never | 165 | 96.8% | 95.7% | 97.8% | 97.5% | 96.2% |
| Former | 187 | 94.9% | 93.1% | 96.5% | 96.1% | 93.8% |
| Current | 198 | 94.5% | 92.9% | 96.0% | 95.6% | 93.5% |

**Performance by FEV1/FVC Range (Disease Severity)**

| FEV1/FVC Range | Severity | N | Accuracy | Sensitivity | Specificity | PPV | NPV |
|---|---|---|---|---|---|---|---|
| <0.50 | Severe | 78 | 97.4% | 96.2% | 98.7% | 98.4% | 96.8% |
| 0.50 – 0.59 | Moderate | 112 | 96.4% | 95.5% | 97.3% | 97.1% | 95.8% |
| 0.60 – 0.69 | Mild | 98 | 93.9% | 91.8% | 95.9% | 95.5% | 92.5% |
| >=0.70 | Normal | 262 | 95.8% | 94.3% | 97.1% | 96.8% | 94.9% |

*Table 11: Stratified Performance Metrics (n=550 Test Cases)*

## Key findings:

- **Consistent performance** across age groups (94.8%-95.6%)
- **Minimal sex disparity** (Male: 95.4%, Female: 95.0%)
- **Highest accuracy** in normal BMI category (96.1%)
- **Best performance** in never smokers (96.8%)
- **Highest accuracy** in severe COPD cases (97.4%)
- **Lower accuracy** in mild COPD cases (93.9%) due to diagnostic uncertainty

This format separates each demographic category into its own table, making it much easier to read and understand the performance across different patient subgroups.

## APPENDIX C: DATASET CHARACTERISTICS

This appendix provides detailed statistics on the training and test datasets used to develop and validate the COPD Prediction System.

### C.1 Chest X-Ray Image Dataset

**Training Set:**

- Total images: 4,249
- Emphysema class: 2,124 images (50.0%)
- Normal class: 2,125 images (50.0%)
- Image format: JPEG
- Resolution range: 256×256 to 2048×2048 pixels
- Preprocessing: Resize to 128×128, normalize to [0,1], RGB conversion

**Validation Set:**

- **Total images: 472**
- **Emphysema class: 236 images (50.0%)**
- **Normal class: 236 images (50.0%)**

**Test Set:**

- **Total images: 550**
- **Emphysema class: 275 images (50.0%)**
- **Normal class: 275 images (50.0%)**

### C.2 Clinical Spirometry Dataset

**Training Set (n=900):**

| Feature | Mean ± SD | Range |
|---|---|---|
| Age (years) | 53.2 ± 14.8 | 18 - 95 |
| Height (cm) | 168.5 ± 9.2 | 145 - 195 |
| Weight (kg) | 74.3 ± 15.6 | 45 - 125 |
| BMI (kg/m²) | 26.1 ± 5.3 | 16.2 – 42.8 |
| F EV1 (L) | 4.41 ± 1.52 | 1.2 – 7.8 |
| FVC (L) | 5.78 ± 1.48 | 2.1 – 9.5 |
| FEV1/FVC | 0.672 ± 0.182 | 0.28 – 0.95 |

*Table 12: Spirometry Dataset*

**Categorical Features:**

- Sex: Male 52%, Female 48%
- Smoking Status: Never 33%, Former 37%, Current 30%
- COPD Label: Positive 50%, Negative 50%

**Test Set (n=100):**

- Similar distribution to training set
- Balanced COPD/Normal split (50/50)
- Representative demographic coverage

## APPENDIX D: GLOSSARY OF TERMS

This glossary is here to help people understand the technical terms that are used in this report. It is meant to assist readers who come from backgrounds.

1. **ANN (Artificial Neural Network):**
   The Artificial Neural Network is a model that is inspired by the biological neural networks in our brain. It is made up of nodes also known as neurons that are organized in layers. In this project the Artificial Neural Network is used to process features to predict COPD.
2. **AUC-ROC (Area Under the Receiver Operating Characteristic Curve):**
   The Area Under the Receiver Operating Characteristic Curve is a way to measure how well a binary classification model is performing. It looks at how the model can tell the difference between positive and negative classes. The values range from 0.5, which's like guessing randomly to 1.0, which is perfect.
3. **BMI (Body Mass Index):**
   The Body Mass Index is a measure of how fat a person has based on their height and weight. It is calculated by taking the persons weight in kilograms and dividing it by their height in meters squared. The categories are Underweight if the BMI is than 18.5 Normal if it is between 18.5 and 25.0 Overweight if it is between 25.0 and 30.0 and Obese if it is 30.0 or more.
4. **CNN (Convolutional Neural Network):**
   The Convolutional Neural Network is a type of deep learning architecture that's good at processing data that is in a grid like images. The Convolutional Neural Network uses layers to automatically learn features from the data. In this project the Convolutional Neural Network is used to look at chest X-rays.
5. **COPD (Chronic Obstructive Pulmonary Disease):**
   The Chronic Obstructive Pulmonary Disease is a lung disease that gets worse over time and makes it hard to breathe. It is usually caused by smoking or being around things that're bad for the lungs. The Chronic Obstructive Pulmonary Disease includes emphysema and chronic bronchitis.
6. **Emphysema:**
   The emphysema is a type of Chronic Obstructive Pulmonary Disease that destroys the walls of the air sacs in the lungs. This makes it hard for the lungs to get oxygen and makes the air

spaces bigger. On a chest X-ray emphysema looks like areas that're not as dense and are over-inflated.

7. **FEV1 (Forced Expiratory Volume in 1 second):**

   The Forced Expiratory Volume in 1 second is the amount of air that a person can breathe out in one second after taking a breath. It is measured in liters. A normal range is between 2.0 and 6.0 liters depending on the persons age, sex and height.

8. **FVC (Forced Vital Capacity):**

   The Forced Vital Capacity is the total amount of air that a person can breathe out after taking a full breath. It is measured in liters. A normal range is between 2.5 and 6.5 liters depending on the persons age, sex and height.

9. **FEV1/FVC Ratio:**

   The FEV1/FVC Ratio is the ratio of the Forced Expiratory Volume in 1 second to the Forced Vital Capacity. It is a way to measure if the airways are blocked. If the ratio is than 0.70 it may mean that the person has Chronic Obstructive Pulmonary Disease according to the Global Initiative for Chronic Obstructive Lung Disease guidelines.

10. **GOLD (Global Initiative for Chronic Obstructive Lung Disease):**

    The Global Initiative for Chronic Obstructive Lung Disease is an organization that makes guidelines for diagnosing and treating Chronic Obstructive Pulmonary Disease. They say that if the FEV1/FVC Ratio is than 0.70 it means that the person has airflow obstruction.

11. **Grad-CAM (Gradient-weighted Class Activation Mapping):**

    The Gradient-weighted Class Activation Mapping is a way to explain how a Convolutional Neural Network makes predictions. It highlights the parts of the image that're most important for the prediction.

12. **HRCT (High-Resolution Computed Tomography):**

    The High-Resolution Computed Tomography is a type of CT scan that takes detailed pictures of the lungs. It is better than a chest X-ray at finding emphysema.

13. **Keras:**

    Keras is a tool that makes it easier to build and train deep learning models. It runs on top of TensorFlow.

14. **NPV (Negative Predictive Value):**

    The Negative Predictive Value is the chance that a person who tests negative really does not have the disease. It is calculated by looking at the number of negatives and false negatives.

**15. PPV (Positive Predictive Value):**

The Positive Predictive Value is the chance that a person who tests positive really has the disease. It is calculated by looking at the number of positives and false positives.

**16. Sensitivity (Recall, True Positive Rate):**

The Sensitivity is the proportion of people who really have the disease and test positive. It is calculated by looking at the number of positives and false negatives.

**17. SHAP (SHapley Additive exPlanations):**

The SHapley Additive exPlanations is a way to explain how a model makes predictions. It looks at how much each feature contributes to the prediction.

**18. Specificity (True Negative Rate):**

The Specificity is the proportion of people who really do not have the disease and test negative. It is calculated by looking at the number of negatives and false positives.

**19. Spirometry:**

The Spirometry is a test that measures how well the lungs are working. It is the way to diagnose and monitor Chronic Obstructive Pulmonary Disease.

**20. StandardScaler:**

The StandardScaler is a way to get all the features to have the scale. It makes sure that features that have numbers do not have too much influence on the model.

**21. Streamlit:**

The Streamlit is a tool that makes it easy to build web applications that're interactive. It is good, for data science and machine learning projects.

**22. TensorFlow:**

The TensorFlow is a tool that makes it easy to build and train deep learning models. It is made by Google. Is very powerful.

# CAPSTONE PROJECT CERTIFICATE


# GITHUB UPLOAD SCREENSHOT